

Deep learning-based individual tree crown delineation in mangrove forests using very-high-resolution satellite imagery

Guillaume Lassalle^{a,*}, Matheus Pinheiro Ferreira^b, Laura Elena Cué La Rosa^c, Carlos Roberto de Souza Filho^a

^a Geosciences Institute, University of Campinas, PO Box 6152, 13083-855 Campinas, SP, Brazil

^b Cartographic Engineering Department, Military Institute of Engineering (IME), Praça Gen. Tibúrcio 80, 22290-270 Rio de Janeiro, RJ, Brazil

^c Department of Electrical Engineering, Pontifical Catholic University of Rio de Janeiro (PUC), R. Marquês de São Vicente 225, 22541-041 Rio de Janeiro, RJ, Brazil

ARTICLE INFO

Keywords:

Mangrove forest
Tree crown delineation
Satellite imagery
Deep learning
Image segmentation

ABSTRACT

Mangrove forests are vulnerable ecosystems that require broad-scale monitoring. Various solutions based on satellite imagery have emerged for this purpose but still suffer from the lack of methods to accurately delineate individual tree crowns (ITCs). Within-stand variability in crown size and shape, crown clumping and fragmentation, and understory vegetation hamper the delineation in these ecosystems. To cope with these factors, the proposed method combines a deep learning-based enhancement of ITCs with a marker-controlled watershed segmentation algorithm. The MT-EDv3 neural network is employed to compute the normalized Euclidean distance of crown pixels to treetops and a Laplacian of Gaussian filter is applied to the resulting image to enhance crown borders before segmentation. The method was applied to WorldView imagery over four mangrove sites worldwide and compared to previously published methods using standardized metrics. Accurate detection (Overall Accuracy ≥ 0.93 and Kappa ≥ 0.87) and area estimation ($R^2 \geq 0.66$, NRMSE $\leq 12\%$) of crowns was achieved for all sites using either the panchromatic band or a combination of the pan-sharpened visible-near-infrared bands. Based on Precision, Recall, and F1-score, the proposed method outperformed previous watershed segmentation and software-based algorithms of crown delineation, as well as the Mask R-CNN segmentation framework. The viewing geometry of images and the forest heterogeneity were identified as important contributors to the delineation accuracy. This study is the first to achieve accurate delineation of ITCs in mangrove forests across sites, opening perspectives of applications to satellite-based monitoring. The method shows promising transferability to other very-high-resolution satellite sensors as well as to aerial and unmanned aerial vehicle imagery and could be improved by including more spectral information and LiDAR-derived canopy height models.

1. Introduction

Mangroves are a type of wetland ecosystem found in the intertidal zone along tropical and subtropical coastlines and estuaries (Lewis et al., 2011; Wang et al., 2019). They are formed by dense forests, which shelter a wide diversity of living organisms and provide more than 20 ecological services and 45 natural products (Barbier, 2016; Hogarth, 2015; Lewis et al., 2011; Vo et al., 2012). Because they colonize the transition zone between terrestrial and marine environments, mangroves act as “coastal kidneys” by limiting land contaminant transport and as natural barriers against waves (Duke, 2016). More importantly, their carbon production rate (up to 700 Tg C.year⁻¹) equals that of

tropical forests, making mangrove ecosystems one of the most valuable contributors to the global blue carbon pool (Alongi, 2012).

Although mangrove forests currently cover more than 137 000 km² in 118 countries, nearly 35% disappeared since the end of the 20th century due to natural and anthropogenic factors (Bunting et al., 2018; Duke et al., 2007; Thomas et al., 2017). Floods, storms, and hurricanes are strong drivers of mangrove loss, but about 60% of the decrease is attributed to human activities according to the 2021 *State of the World's Mangroves* report (Spalding and Leal, 2021). In particular, human-induced mangrove loss falls into four categories according to Makowski and Finkl (2018): resource exploitation (e.g., wood, oil, medicine), competition for space (aquacultural, urban, and industrial

* Corresponding author at: University of Campinas, UNICAMP, PO Box 6152, 13083-855 Campinas, SP, Brazil.

E-mail address: guillaumelassalle.pro@gmail.com (G. Lassalle).

<https://doi.org/10.1016/j.isprsjprs.2022.05.002>

Received 6 November 2021; Received in revised form 4 May 2022; Accepted 13 May 2022

Available online 25 May 2022

0924-2716/© 2022 The Authors. Published by Elsevier B.V. on behalf of International Society for Photogrammetry and Remote Sensing, Inc. (ISPRS). This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

development), human modification (freshwater resource diversion) and pollution (oil spills, wastewater), and management failure (*i.e.*, failures in planning policies and ecosystem evaluation). In addition, sea-level rise, global warming, genetic isolation, and pests are becoming severe threats to mangroves, which are now among the most preoccupying ecosystems worldwide even though they represent less than 0.5% of the global forest coverage (Duke, 2016; Krauss et al., 2014; Ward et al., 2016). Therefore, monitoring mangrove forests is critical to maintaining their invaluable services in the long term.

Because mangrove trees form dense canopies with limited field access, remote sensing (RS) proved a reliable solution to study these forests at a large scale. According to recent reviews (Heumann, 2011; Wang et al., 2019), attention was mostly given to forest extent mapping, species classification, biophysical-biochemical parameter retrieval, and carbon stock estimation over the past 65 years. More recently, health assessment and stress monitoring applications have emerged as higher spatial and spectral resolutions gave access to finer information on tree characteristics (Hati et al., 2020; Kumar et al., 2020). Passive airborne and satellite optical RS has been primarily employed to monitor mangrove forests, including low (MODIS), medium (Landsat, Sentinel-2), and high and very-high (SPOT, QuickBird, WorldView) spatial resolution imagery (Lagomasino et al., 2014; Manna and Raychaudhuri, 2020; Pastor-Guzman et al., 2018; Valderrama-Landeros et al., 2018). Besides, a substantial number of studies have emphasized the use of LiDAR as a valuable tool for mangrove mapping and monitoring (Li et al., 2021; Zhang, 2008). Very-high resolution (VHR) imagery is best suited to map mangrove species, phenology, and to retrieve canopy biochemistry accurately, as it requires identifying trees individually beforehand (Aval et al., 2018; Ferreira et al., 2018). Individual Tree Crown (ITC) delineation is more important in forest health assessment as the response of trees to stressors and disturbances varies among species and between individuals from a single species. Consequently, automatic ITC delineation benefits airborne and satellite monitoring of forests. However, it remains poorly documented on mangroves, not due to a lack of interest, but mainly because achieving accurate delineation of tree crowns is usually more challenging in mangroves than in other forest types.

ITC delineation consists of grouping pixels from the same tree crown into a single image object. It is usually achieved by applying segmentation algorithms to aerial and satellite optical images, or LiDAR-derived Canopy Height Models (CHMs) (Bunting and Lucas, 2006; Chemura et al., 2015; Johansen et al., 2020; Xu et al., 2021; Yun et al., 2021). One critical requirement to accurately delineate ITCs is that pixel size must be smaller than the crown size, so that meter (1–2 m) to sub-meter resolution is often recommended. Although there is no consensus on the best method to delineate ITCs, watershed, valley-following, and region-growing segmentation are among the most widely used algorithms (Ke and Quackenbush, 2011a). They usually perform well in boreal and temperate forests with bright treetops, regularly shaped tree crowns, and crown shyness (*i.e.*, inter-crown spaces). Conversely, crown separability is highly challenging in mangroves owing to (i) the presence of understory vegetation (mangrove saplings, shrubs, and saltmarsh vegetation) filling inter-crown spaces, (ii) the fragmentation of crowns; *i.e.*, the decomposition of a crown into several sub-crowns (multi-stem trees), and (iii) the clumping and overlapping of crowns from two or more trees (Li et al., 2021; Yin and Wang, 2019). In addition, mangrove tree development – and therefore crown morphology – varies among species and is influenced by the growing conditions (salinity, nutrient availability, pollution) and light availability, inducing high variability among crowns within a single forest patch (Krauss et al., 2008; Lewis et al., 2011; Vovides et al., 2018). All these reasons explain the lower performance of standard ITC delineation methods in mangrove forests compared to other forest types.

One possible way to facilitate ITC delineation is to enhance crown edges by preprocessing images before segmentation. Band selection, dimensionality reduction, and image filtering are the most efficient ways

to achieve this (Ke and Quackenbush, 2011a; Wagner et al., 2018). Alternatively, recent progress has been made in deep learning-based ITC delineation, by convolutional neural networks (CNNs) to segment images or enhance treetop detection and pixel homogeneity within crowns. For example, Martins et al. (2021) delineated crowns and classified species of urban trees by applying a multi-task CNN to RGB aerial images, while Braga et al. (2020) and Hao et al. (2021) applied the Mask R-CNN developed by He et al. (2017) for tree detection and ITC delineation in an Atlantic rainforest and tree plantations, respectively. However, no application to mangrove forests has been yet proposed. More broadly, little progress has been made in mangrove ITC delineation over the past decades. To the best of our knowledge, only a few site-specific methods have been proposed for this purpose. Wannasiri et al. (2013) delineated mangrove trees with 87% accuracy by applying a variable window filtering to LiDAR data in Thailand. Yin and Wang (2019) achieved 46% accuracy by segmenting a LiDAR-derived CHM in mangroves from China surveyed with an Unmanned Aerial Vehicle (UAV). This study was recently followed by Qiu et al. (2019), who successfully applied the same method. Although LiDAR-based ITC delineation is promising, it is still limited to small-scale applications. As suggested by several authors, it would benefit from being combined with very-high-resolution satellite imagery. Heenkenda et al. (2015) achieved 92% accuracy in delineating Australian mangrove trees using WorldView-2 imagery, by implementing a local maximum growing method that combines the panchromatic, red, and NIR1 (765–901 nm) bands in the multiresolution segmentation algorithm. The latter was also applied by Kamal et al. (2016) to segment WorldView-2 imagery in Indonesian and Australian mangroves, but no accuracy assessment was performed on the resulting delineation. In the study of Hirata et al. (2014), QuickBird images were segmented to delineate uniform mangrove patches rather than ITCs in Thailand. Hence, although the conservation of mangroves is becoming a growing concern worldwide, there is still a lack of solutions to monitor these forests across sites, at tree level, and at large scale, despite the availability of VHR satellite imagery.

This study examines the feasibility of delineating ITCs in mangrove forests at large scale by applying a deep learning-based method to VHR satellite imagery. More specifically, the following questions are addressed: (i) Are VHR satellite images suitable to delineate mangrove ITCs accurately across sites? (ii) What spectral bands perform best to achieve such a delineation? (iii) How does the proposed method perform compared to the commonly used ITC delineation methods? We tested our method across four mangrove forests from different regions of the globe, all varying in tree configuration (clumped, boundary, and isolated trees), species composition, crown size and shape, and understory vegetation density. The manuscript is organized as follows. Section 2 first describes the selected study sites and the satellite images used. Then, a comprehensive description of the ITC delineation method is provided. This section ends by detailing the assessment of the method's performance. Section 3 describes the results obtained for the four studied sites and compares the proposed method's results to other ITC delineation methods. Finally, the sources of inaccuracies and the perspectives of application are discussed in Section 4.

2. Materials and methods

2.1. Study sites

The study was conducted on four selected mangrove stands (Fig. 1, see also Supplementary Fig. S1). The first is a degraded fringe mangrove of approximately 330 ha bordering the Bertioga channel, Brazil (23°54'09.3"S, 46°12'18.4"W). Only three 'true' mangrove species are found on this site, and the understory vegetation is composed of sparse to dense saplings (Lamparelli et al., 1997). Tree height rarely exceeds 10 m in this forest (Soares and Schaeffer-Novelli, 2005), and crown fragmentation and clumping are commonly observed. The second site is located in the Everglades National Park (United States). A dense,

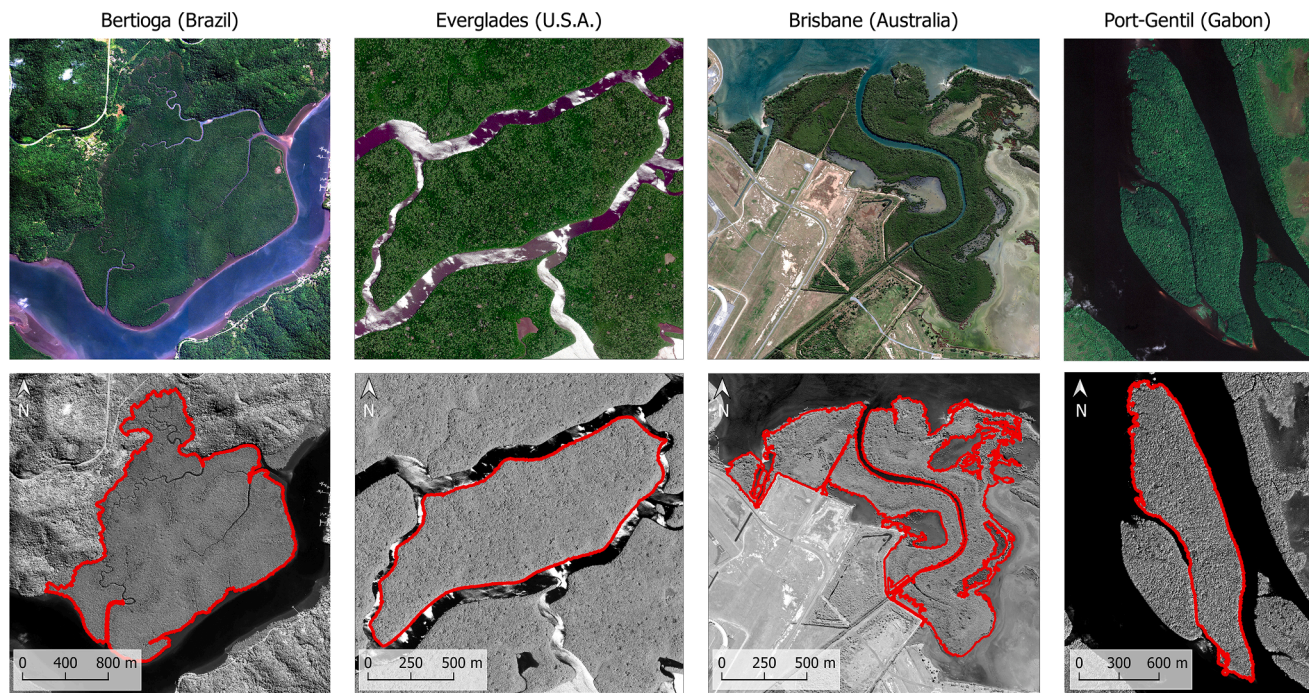


Fig. 1. Overview of the study sites as seen by WorldView-3 (Brazil, U.S.A., Australia) and WorldView-4 (Gabon) satellite imagery. (Upper figures: PAN-sharpened RGB true color composite images; Lower figures: Panchromatic image). The limits of the selected mangrove forests are marked by red polygons. (See also Supplementary Tab. S1). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

riverine mangrove stand of 110 ha, with tree height ranging from 6 to 24 m, was selected within the park (25°21'45.7"N, 81°04'10.0"W). This mangrove forest includes the following species: red (*Rhizophora mangle* (L.)), black (*Avicennia germinans* (L.)), and white (*Laguncularia racemosa* (L.)) mangrove, as well as button mangrove (*Conocarpus erectus* L.) (Castañeda-Moya et al., 2011). The third mangrove stand is located in Moreton Bay near Brisbane, Australia (27°21'55.9"S, 153°08'36.5"E). This forest of 115 ha is dominated by white mangrove (*Avicennia marina* (Forssk.) Vierh.), but six different species are found in the region (Sun et al., 2014). Vegetation height ranges from 2-m shrubs to 30-m trees, forming dense patches with overlapping crowns (Kamal et al., 2014). The last site is a mangrove forest (90 ha) from the Gabon estuary located near the Port-Gentil seaport (0°47'02.4"S, 8°57'39.6"E). This region is home to the tallest mangrove trees globally, reaching up to 63-m height and being dominated by the red mangrove species *Rhizophora racemosa* G. Mey. (Aldous et al., 2021; Simard et al., 2019). As described here, the four sites differ in the morphology of the upper canopy and the density of understory vegetation strata. Therefore, they were well-suited to assess the robustness of the proposed ITC delineation method.

2.2. WorldView imagery

VHR imagery is recommended to achieve an accurate delineation of mangrove ITCs. In this study, we used level-2A (surface reflectance) WorldView-3 and -4 products provided by MAXAR (Westminster, CO, USA). The images were acquired over the four study sites between August 2016 and August 2019, under cloud-free conditions, with off-nadir viewing angles ranging from 11.5 to 17° (see Supplementary Tab. S1). Depending on the multispectral instrument, the images included four (WorldView-4) to eight (WorldView-3) bands over the Visible Near-Infrared (VNIR) spectral domain at 1.2 m spatial resolution and a panchromatic band (PAN) at 0.3 m. To enhance the resolution of images, the VNIR bands were pan-sharpened using the Nearest Neighbor Diffusion (NNDiffuse) algorithm available in ENVI® (L3Harris Geospatial, Boulder, CO, USA) (Sun et al., 2014). NNDiffuse is based on spectral mixture analysis concepts and assumes that the spectrum of a

pan-sharpened image pixel can be modeled using the spectral response of its neighboring pixels. In our case, it proved best suited than other commonly used pan-sharpening algorithms (see Supplementary Fig. S2). NNDiffuse requires the user to set some parameters such as pixel size ratio (between the multispectral and PAN bands), spatial smoothness, and intensity smoothness. In our case, we set the pixel size ratio to 4, spatial smoothness to 2.48, and intensity smoothness was dynamically adjusted to local similarity. The resulting pan-sharpened images have 0.3-m of pixel size, eight bands for WorldView-3 and four bands for WorldView-4 (Fig. 1). Thus, both the spectral and the very-high spatial information was exploited in the proposed method to delineate ITCs.

2.3. Individual tree crown dataset

Deep learning-based delineation of ITCs usually requires a large dataset of well-identified crowns. To create this dataset, we manually delineated a total of 2000 mangrove ITCs over the four study sites (500 ITCs per site), selecting isolated, clumped, and boundary trees with varying crown sizes (<50 to >4000 pixels) and shapes (e.g. compact, loose, round, elliptic). The delineation was carried out by a single person using the PAN and VNIR bands alternatively to clearly distinguish the crown borders and minimize errors in the resulting crown polygons. For each study site, the manually delineated ITCs were randomly split into 70% for training ($n = 350$ ITCs) and 30% for testing ($n = 150$ ITCs). The distribution of crown areas, as well as the crown statistics for the training and testing ITCs, are shown in Fig. 2. For a complete view of ITC polygons, the reader is referred to Supplementary Fig. S3-S6. The ITCs strongly differed in area and shape among the four sites (Fig. 2), making it challenging to develop an effective, across-site delineation method.

2.4. Proposed method

The proposed method relies on two successive steps: (1) crown separability enhancement and (2) automatic delineation of ITCs (Fig. 3). In the first step, a multi-task CNN is used to generate a distance map (Fig. 3a-b), in which the pixels inform the distance to the ITC

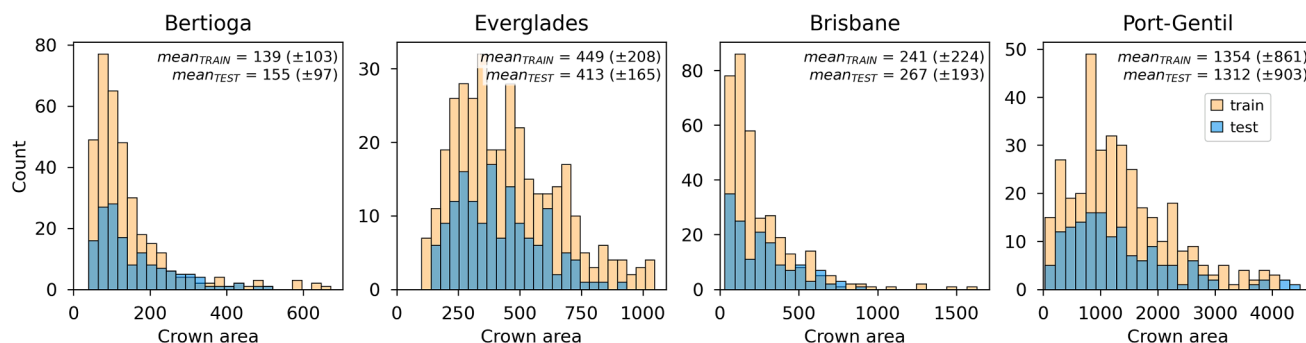


Fig. 2. Distribution of the manually delineated crown area for the train ($n = 350$) and test ($n = 150$) datasets over the four study sites. For each dataset, bars represent the count of crowns for a given crown area. Crown area is given in 0.3-m pixel unit.

boundaries. Then, a Laplacian of Gaussian (LoG) filter is applied to the distance map to enhance the crown borders and the treetop (Fig. 3c). In the second step, ITCs are automatically delineated using a marker-controlled watershed segmentation algorithm (Fig. 3d). In the following sections, we detail the proposed method.

2.4.1. Crown separability enhancement

The first level of crown separability enhancement consists in producing a distance image in which the pixel values represent the distance to the ITC border. To obtain it, we used the DeepLabv3+ based Multi-Task Encoder-Decoder (MT-EDv3) network proposed by La Rosa et al. (2021). This network was developed to perform ITC delineation and species classification and was successfully employed for tree species mapping in urban and forest environments (Martins et al., 2021; La Rosa et al., 2021). The MT-EDv3 is a multi-task network designed to jointly perform semantic segmentation and distance map estimation from an input image. Specifically, the network consists of a shared encoder that learns relevant representations for both tasks, and two decoders that learn task-specific representations. The shared encoder is based on the ResNet network (He et al., 2016), whereas the decoder for the semantic segmentation task is based on DeepLabv3+ architecture (Chen et al., 2018). The decoder in charge of estimating the distance map was implemented using traditional convolutions and bilinear upsampling. The well-known focal loss function was used to train the semantic segmentation branch, and the mean square error was selected as the loss function for the distance map estimation branch. The potential of using the distance map for ITC detection and shape delineation was highlighted in La Rosa et al. (2021). Because classifying mangrove species was out of the scope of our study, the semantic segmentation branch was used as a secondary task to improve the generalization of the distance map estimation.

MT-EDv3 was trained with three input images: (i) the PAN band or a NIR-red-green band composition of the pan-sharpened VNIR image (Fig. 4a), (ii) a reference binary image obtained by a vector-to-raster conversion of training ITC polygons ($n = 350$ crowns per site, Section 2.3) (Fig. 4b) and (iii) a reference distance map obtained by computing the normalized Euclidean distance transform between each pixel and the border of the training ITCs (Fig. 4c). As described in Martins et al. (2021) and La Rosa et al. (2021), the network is trained using the partial loss principle (Lin et al., 2017), which consists in calculating the loss function using only the pixels from the annotated ITCs (white pixels in Fig. 4b) and ignoring the unknown region (black pixels in Fig. 4b). This technique alleviates the need for a very large set of annotated ITC samples to achieve a good performance. Once trained with the reference distance map, the MT-EDv3 was applied to the entire PAN and VNIR images to produce a distance map with all potential ITCs present in the mangrove forest (Fig. 4d). For more information about the network architecture and hyperparameters configuration, we refer the reader to Martins et al. (2021).

The training process begins by extracting 20 000 image patches (18

000 for training and 2000 for validation), ensuring that at least 10% of the patch contains a reference ITC. For each site, we chose a different patch size according to the average crown area of the site (Fig. 2). Thus, the patch size was set to 64×64 pixels for Bertioga, 128×128 pixels for the Everglades and Brisbane, and 256×256 pixels for Port-Gentil. Consequently, crown area represented 1.5 to 3.4% of the patch area on average depending on the site, so each patch encompassed several crowns, including the largest ones (>4000 pixels in Port-Gentil). We trained the model with both the PAN band and VNIR color composition, separately.

Network generalization was improved via data augmentation by implementing random rotations (90 , 180 , and 270°), and horizontal and vertical flips to the patch images. To avoid overfitting, a dropout rate of 0.65 was set. The stochastic gradient descent optimizer (Murphy, 2012) with a momentum of 0.9 and an initial learning rate of 0.05 was used. The model was trained for 20 epochs with a batch size of 8. At each epoch, a new set of training and validation patches were extracted. Predictions were performed using a sliding window with 10, 30, and 50% overlapping rates and computing the mean over the three outputs. The MT-EDv3 network was trained on a computer equipped with two Intel® Xeon Silver processors (2.40 GHz), 128 GB RAM, and a NVIDIA® RTX 2070 SUPER graphic card with 8 GB memory, taking approximately 1–5 h to complete (20 epochs) depending on the overall forest extent. Codes were implemented in Python language using TensorFlow and CUDA® 11.4 and can be made available by authors upon reasonable request.

The reconstructed distance map produced by the MT-EDv3 network, which resulted from the merging of predicted distance map patches, was smoothed using a 2D Gaussian filter with σ values set to 1.5 (Bertioga and Everglades) or 2 (Brisbane and Port-Gentil). These values were selected as they resulted in a seamless distance map while preserving the crown edges. At this stage, mangrove trees could be easily identified on the image. However, the separability of crown borders was still unsatisfactory and needed further improvements towards achieving an accurate delineation of ITCs. To that aim, we applied a LoG filter to the smoothed distance image (Marr and Hildreth, 1980). Edge detection filters already proved relevant to contrast crown borders in previous studies (Pouliot et al., 2002; Wagner et al., 2018; Wang et al., 2004). The LoG filter first applies a Gaussian smoothing to remove noise from the image, which had already been done in our case as mentioned above. Then, the algorithm finds zero crossings in the second derivative of the smoothed images, which correspond to the crown edges. Here, the LoG was applied to the smoothed distance images with a kernel size of 9×9 pixels. Preliminary tests showed that smaller kernel sizes did not enhance the difference between crown edges and non-crown pixels, while larger kernels tend to merge the edges of neighboring crowns. The resulting edge-enhanced images served for delineating ITCs (Fig. 3c-d).

2.4.2. Automatic delineation of ITCs

Mangrove ITCs were automatically delineated from the edge-

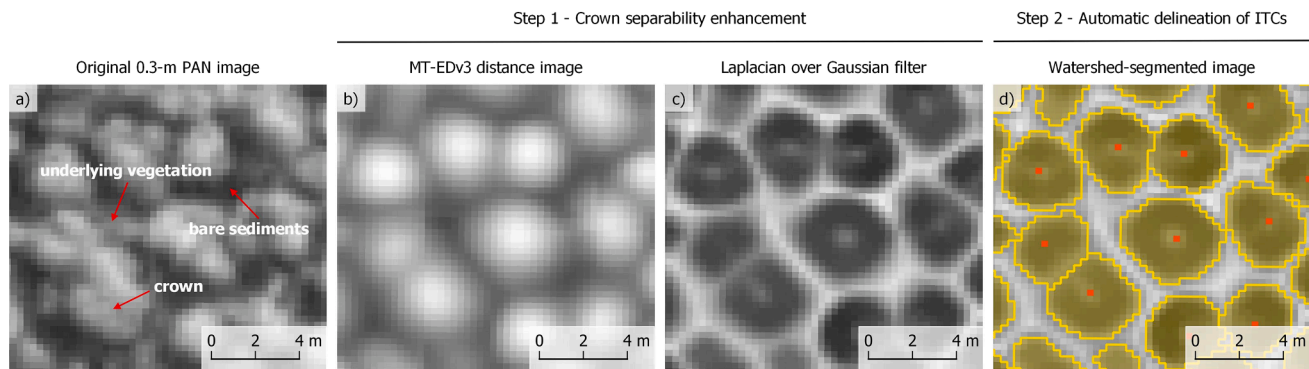


Fig. 3. Example of the proposed individual tree crown (ITC) delineation method on a WorldView-3 panchromatic image (pixel size = 0.3 m) acquired over the Bertioga mangrove site. (a) Original panchromatic image in which the ITCs are not clearly seen. (b) Distance map obtained by the MT-EDv3 convolutional neural network (see Fig. 4 for details). In this image, the pixel values represent the distance to the ITC boundary. (c) Result of the application of the Laplacian of Gaussian (LoG) filter to enhance crown borders and treetops. (d) Result of ITC delineation performed by a marker-controlled watershed segmentation algorithm. The yellow polygons and red squares denote the segmented individual tree crowns and the treetops, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

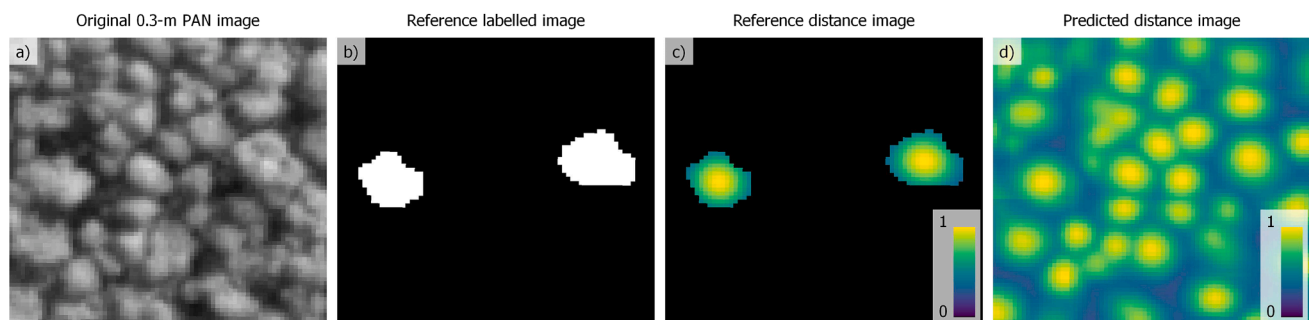


Fig. 4. Example of application of the MT-EDv3 network on a WorldView-3 PAN image. Training crowns are manually delineated from the original image (a) to produce a reference labeled image (b) (white: Training crowns, black: Background) and a reference distance image (c). Then, all these images are used to train the MT-EDv3 network before applying it to the entire original image to produce the final distance image (d).

enhanced images (Fig. 3c) over the four study sites using a marker-controlled watershed segmentation (Huang et al., 2018; Tong et al., 2021). Watershed segmentation usually results in over-segmentation owing to image noise, which, in our case, would have resulted in the splitting of a single mangrove crown into several sub-crowns. The marker-controlled watershed segmentation copes with this issue by imposing the number and location of “basin outlets”, here represented by treetops. Here, treetops were detected by applying a local maximum detection filter to the Gaussian-smoothed distance image, assuming that treetops are located near or at the center of crowns (the most distant from the crown edges, *i.e.* normalized Euclidean distance ≈ 1). Finally, the delineation of ITCs was achieved by applying the watershed segmentation algorithm to the LoG-filtered image, using the detected treetops as markers and the enhanced crown borders as “watershed ridge lines” (Fig. 3d). The method was applied separately to the PAN and VNIR (NIR-red-green) images and the accuracy of ITC delineation was assessed as described in the following section.

2.5. Assessment

A multi-criteria assessment of the delineation was conducted separately for each site, involving multiple, complementary metrics, as proposed by Yin and Wang (2016). The first assessment level consisted of determining whether the method succeeded in distinguishing crowns from non-crowns areas, the latter encompassing bare areas, inter-crown spaces, water, and understory vegetation (Braga et al., 2020). For this purpose, 1000 validation points were randomly generated over each study site, so that half of them fell in crown pixels and the remaining half

in non-crown pixels, the latter encompassing bare areas, inter-crown spaces, water, and understory vegetation. The validation points were then compared to the predicted ITCs through confusion matrices. The Overall Accuracy (OA) and Cohen’s Kappa coefficient of the predictions were determined (Cohen, 1968; Story and Congalton, 1986). Here, false negatives indicated visually interpreted crown pixels that did not match with any predicted crown (omission errors). In contrast, false positives corresponded to non-crown pixels intersected by predicted crowns (commission errors).

We compared crown intersections between the true (manually-delineated) and predicted ITCs. For a given true crown, we counted the number of predicted crowns intersected. The process was repeated for all the true crowns ($n = 150$ ITCs per site), and the frequency of intersection (one, two, or three and more predicted crowns) was determined as described in Wagner et al. (2018a). If most of the true crowns are well-predicted, they will intersect with a single predicted crown. Conversely, if true crowns are oversegmented, they will intersect with two or more predicted crowns.

From all predicted crowns intersecting true crowns, we retained those sharing at least 50% of Intersection over Union (IoU) with the true crowns, which are usually considered as correctly delineated (Tochon et al., 2015). IoU is defined as the bounding box area corresponding to the intersection (or overlap) of a true and predicted crown, divided by the sum (or union) of the true and predicted crown bounding box areas (in pixels). For every ITC with $\text{IoU} > 50\%$, we computed the Precision, Recall, and F1-score (Xu et al., 2021). The Precision is defined as the ratio between the bounding box of the intersection area and that of the predicted crown. Conversely, the Recall is defined as the ratio between

the bounding box of the intersection area and that of the true crown. Both are very sensitive to under- and overestimation of crown areas, with values close to 1 indicating accurate delineation. The F1-score combines both precision and recall:

$$F1 - score = 2 \frac{Precision \times Recall}{Precision + Recall} \quad (1)$$

Readers can refer to the article of [Braga et al. \(2020\)](#) for an illustrated example of the IoU, Recall, and Precision calculation. These metrics were computed for each eligible predicted crown and averaged over the entire study site.

We also performed a more general evaluation of the delineated ITCs. First, the distributions of true and predicted crown areas were compared using a non-parametric Kolmogorov-Smirnov test. According to this test, the absence of significant differences ($P \geq 0.05$) indicates good agreement between the true and predicted ITCs area. To complement this metric, we quantified the quality of predictions by computing the coefficient of determination R^2 , the Root Mean Square Error (RMSE):

$$RMSE = \sqrt{\frac{\sum_{j=0}^n (y'_j - y_j)^2}{n}} \quad (2)$$

with y'_j and y_j being the predicted and true area of crown j , respectively, and the Normalized Root Mean Square Error (NRMSE), defined as:

$$NRMSE = \frac{RMSE}{y_{max} - y_{min}} \quad (3)$$

where y_{max} and y_{min} denote the maximum and minimum crown area in the test set, respectively ([Byrd et al., 2018](#)).

2.6. Comparison with other methods

The deep learning-based delineation of ITCs developed in this study was compared with three previously described methods based on satellite imagery and applied to mangrove forests. These methods were tested on our study sites using the 0.3-m PAN and/or VNIR images as described in the original work. The objective here is to highlight how deep learning can outperform previous delineation methods in mangroves by overcoming some of their limitations, regardless of the forest location.

The first method is a simple marker-controlled watershed segmentation ([Biswas et al., 2020](#); [Tong et al., 2021](#)). However, unlike our method, no pre-enhancing of crowns was applied to the image. Treetops were detected on the PAN image using a local maximum filter and used in the watershed segmentation as described above. Comparing the results of our method to those obtained with the watershed algorithm allowed us to evaluate the contribution of MT-EDv3 to the ITC delineation in mangrove forests.

Based on [Heenkenda et al. \(2015\)](#), a region-growing segmentation combining the PAN, red, and NIR bands in Ecognition® Developer (Trimble Geospatial, Sunnyvale, CA, USA) was applied to the images. This method detects the local maxima in the PAN and NIR bands, which are considered treetops. Then, an iterative region-growing procedure is performed using the treetops as seeds to grown regions. The process is controlled by a parameter, defined as the ratio of the NIR value of the treetop pixel to that of the neighboring pixels. Thus, the method assumes that treetops are brighter than the rest of the crown in the NIR band. False ITCs are finally removed based on their mean Normalized Difference Vegetation Index (NDVI) and the standard deviation of the red band. Finally, a morphological-based smoothing routine implemented in the software is applied to the remaining ITCs.

The last method selected for comparison involves a bottom-up region merging algorithm known as the multiresolution segmentation. This method has been widely used to delineate ITCs from satellite imagery in previous works, including in mangroves ([Johansen et al., 2020](#); [Kamal](#)

[et al., 2016](#); [Singh et al., 2015](#)). Briefly, the algorithm combines one or several bands to create image objects (*i.e.*, ITCs) with minimized heterogeneity. Single-pixel objects are first created and then merged to form larger objects. Their size is controlled by a scale parameter. In addition, the segmentation is controlled by a shape parameter that determines the influence of the spectral information on the segmentation (high shape value = low influence of reflectance), and a compactness parameter, defined as the ratio between the ITC border and area. In this study, scale values ranging from 10 to 100 with an increment of 10 were tested, along with shape fixed to 0.1 (strong influence of reflectance), and compactness ranging from 0.3 to 0.7 with an increment of 0.1. The wide range of shape values was defined to account for the variability in crown size across study sites. The multiresolution segmentation was applied to the original four (WorldView-4) and eight (WorldView-3) VNIR, pan-sharpened images. The green, red, and NIR bands were given more weight than other bands. The optimal set of parameters was selected based on the F1-score.

Finally, our method was compared to another deep learning-based segmentation framework called the Regional Convolutional Neural Network (Mask R-CNN) ([He et al., 2017](#)). Although this framework has never been applied to mangroves, it achieved accurate delineation of ITCs in other forest types ([Braga et al., 2020](#); [Hao et al., 2021](#); [Yu et al., 2022](#)). The Mask R-CNN first detects a target object, draws its bounding box, and then, performs a segmentation of the object encompassed in this box. A comprehensive description of the Mask R-CNN framework and its use in remote sensing of forests – including ITC delineation – can be found in [Chiang et al. \(2020\)](#). In this study, it was applied to detect and delineate mangrove ITCs using the PAN + VNIR composite image and the same patch sizes, data augmentation strategy, and input parameters (epochs, batch size, etc.) as the MT-EDv3 network. Application of the Mask R-CNN was performed in ArcGIS® Pro (ESRI, Redlands, United States), which includes a faithful implementation of the original framework adapted to RS data.

These four methods (marker-controlled watershed, region-growing, multiresolution segmentation, and Mask R-CNN) were finally compared with the proposed method in terms of Precision, Recall, and F1-score computed on the test set of manually delineated ITCs.

3. Results

3.1. Crown recognition

As seen in [Table 1](#), the delineation of ITCs obtained by the proposed method succeeded in distinguishing the crown from the non-crown areas. All sites exhibited $OA \geq 0.93$ and Kappa coefficients ≥ 0.87 , indicating a good agreement between visually generated and predicted crowns. That is, more than 470 out of the 500 crown or non-crown pixels were correctly labeled over the four study sites (see confusion matrices in Supplementary Tab. S2). Because of the minor differences observed between the PAN and VNIR results, no recommendations can be made on the most appropriate bands. However, applying the method to the PAN image resulted in slightly more false-positives. In contrast, the VNIR-derived delineation led to more false-negatives in the Everglades,

Table 1

Overall Accuracy (OA) and Kappa coefficients computed by comparing the visually-generated validation points and the ITC delineation model's predictions using the PAN and VNIR bands, for the four study sites. The corresponding confusion matrices are detailed in Supplementary Tab. S2.

	PAN		VNIR	
	OA	Kappa	OA	Kappa
Bertioga	0.95	0.92	0.95	0.92
Everglades	0.95	0.91	0.93	0.87
Brisbane	0.95	0.91	0.97	0.95
Port-Gentil	0.97	0.95	0.95	0.90

Brisbane, and Port-Gentil mangroves. Usually, in ITC delineation from satellite imagery, false-negatives are attributed to an underestimation of the crown area, *i.e.*, crown pixels not included in the crown segment, whereas false-positives result from an overestimation of the crown area, *i.e.*, the inclusion of non-crown pixels in the crown segment. In mangroves, false-positives may be attributed to understory vegetation (mangrove saplings, saltmarsh vegetation) in inter-crown spaces being poorly distinguished from the crown border, therefore blurring the crown contour. This phenomenon is less likely to be observed with the VNIR image as the NIR band is highly discriminative of mangrove seedlings and mature trees. Although observed here, false-positives and -negatives remain minor using our method as the MT-EDv3 network combined with the LoG filter ensures accurate discrimination of crown and non-crown areas while enhancing the crown contour. The accurate mappings obtained with the four study sites, which differ in many aspects (crown size and shape, understory vegetation, etc.), therefore highlight the robustness of the proposed method and its transferability across mangrove forests.

3.2. Crown intersection

Although the metrics mentioned in Section 3.1 inform the overall quality of the crown mapping, crown intersection provides a better quantitative assessment of the segmentation. Figs. 5 and 6 indicate that the predicted crown polygons matched well with the manually delineated ones, as up to 92% of the latter were intersected by a single polygon (*i.e.*, one predicted crown intersecting a single true crown, also called *one-to-one relationship* (Yin and Wang, 2016)). Yet it must be noticed that the performance varied substantially among sites and according to the spectral bands exploited in the segmentation. The Bertioga PAN image led to the best segmentation since 8% of the true crown were intersected by two or more predicted crowns (Fig. 5). Results were not improved for this site by using the VNIR imagery, as the percentage of predicted crowns intersecting a single crown was reduced from 92% to 68% (Fig. 6). The same applies to the Everglades mangrove, where crowns are about three times larger on average.

Conversely, segmentation quality was not affected by the number of bands employed in the Brisbane site, in which average crown areas ranged between Bertioga and the Everglades. In Port-Gentil, where most crowns exceed 1000 pixels in area, the segmentation was slightly improved by using the VNIR image, reaching 80% of single-crown intersection. Oversegmentation, as indicated by two or more predicted crowns intersecting the true ones, was observed in various proportions for all sites and using both the PAN and VNIR images. However, it remained lower than 32%, indicating an accurate segmentation according to similar work centered on WorldView imagery in other forest types (Braga et al., 2020; Wagner et al., 2018).

Based on Sections 3.1 and 3.2, it can be concluded that the proposed method succeeds in recognizing and segmenting ITCs across mangrove stands. However, assessing the results of delineations cannot be

restricted to classification and intersection metrics as they do not inform on the estimated crown areas. For example, a well-identified (true positive), well-segmented (single-polygon intersect) crown predicted by the method may be smaller or larger than the corresponding true crown. A more quantitative evaluation is therefore recommended.

3.3. Crown area estimation

Considering the predicted crowns with $\geq 50\%$ IoU, crown area was accurately estimated using the proposed method. These results are shown in Figs. 7 and 8 for the PAN and VNIR images, respectively. Based on Kolmogorov-Smirnov tests, no significant difference was observed in the distribution of crown areas between the true and predicted sets for any of the sites and using either the PAN or the VNIR bands ($P \geq 0.05$). The accuracy of the delineation strongly varied among sites and according to the crown area. The R^2 computed between the true vs. predicted crown areas ranged from 0.54 to 0.97. R^2 values above 0.75 were observed while considering only Bertioga, Brisbane, and Port-Gentil mangroves, along with NRMSE not exceeding 10%. The mangrove trees from Port-Gentil exhibited the best crown estimates with only 3% of NRMSE when using the VNIR image as input in the MT-EDv3 network. However, accurate predictions were also observed using the PAN only ($R^2 = 0.83$, RMSE = 8%). A similar trend was observed for the Everglades, where VNIR bands reasonably improved the delineation.

Conversely, ITCs were more accurately delineated using the original PAN image in Bertioga and Brisbane, as indicated by R^2 values ≥ 0.87 and less than 8% of NRMSE. However, for these sites, an underestimation of the true crown area was observed above a certain threshold of about 250 and 500 pixels, respectively. This was confirmed by the distribution of the prediction residuals shown in Figs. 7 and 8.

3.4. Precision, Recall, and F1-score

To better evaluate the delineation of ITCs, taking all possible situations into account (under-, over-, and accurate segmentation), the Precision, Recall, and F1-score were analyzed. Using our method, the averaged Precision and Recall computed between the true and predicted mangrove ITCs exceeded 0.83 for all sites depending on the bands used as inputs (Table 2). There was no substantial difference between these two metrics for a single site, indicating no prevalence of under- or oversegmentation. The accuracy of the delineation was further highlighted by F1-scores > 0.80 . In this regard, the results confirmed the observations made in Section 3.3. Applying the method to the VNIR image increased the F1-score from 0.83 to 0.93 in Port-Gentil, and from 0.81 to 0.85 in the Everglades. Conversely, the PAN image resulted in slightly higher F1-scores (+0.02 to +0.04) than the VNIR in Bertioga and Brisbane. Hence, while avoiding both under- and oversegmentation is a difficult task in the delineation of ITCs, the proposed method proved suitable to accurately recognize and delineate ITCs in mangrove forests of the four study sites.

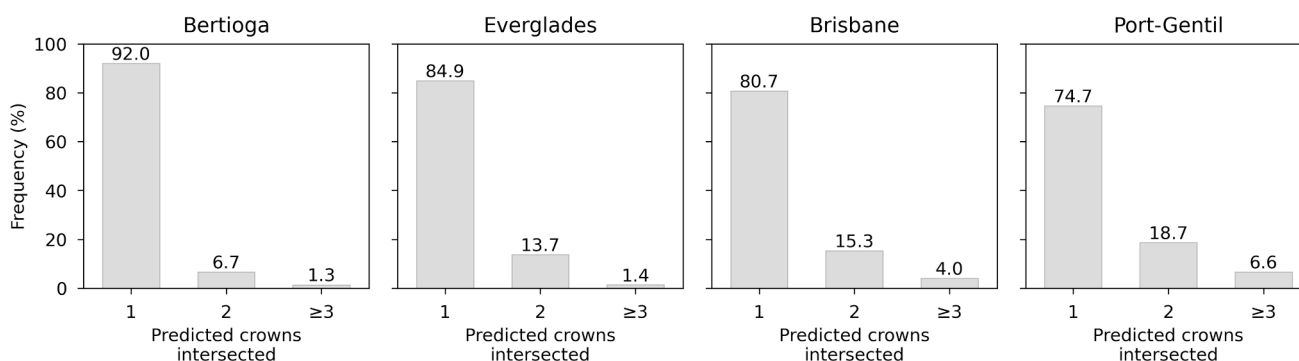


Fig. 5. Frequency of intersection of the true crown polygons with the predicted ones for the four study sites using the PAN band only.

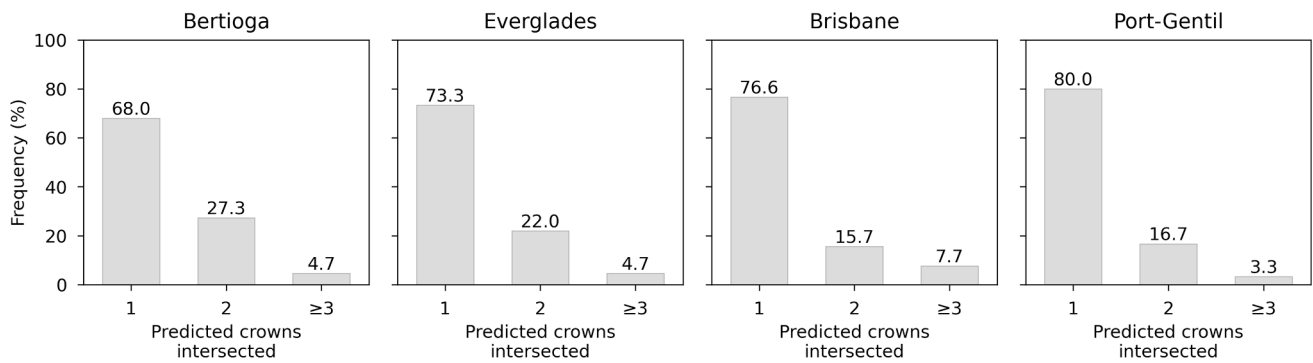


Fig. 6. Frequency of intersection of the true crown polygons with the predicted ones for the four study sites using the VNIR bands (NIR-red-green).

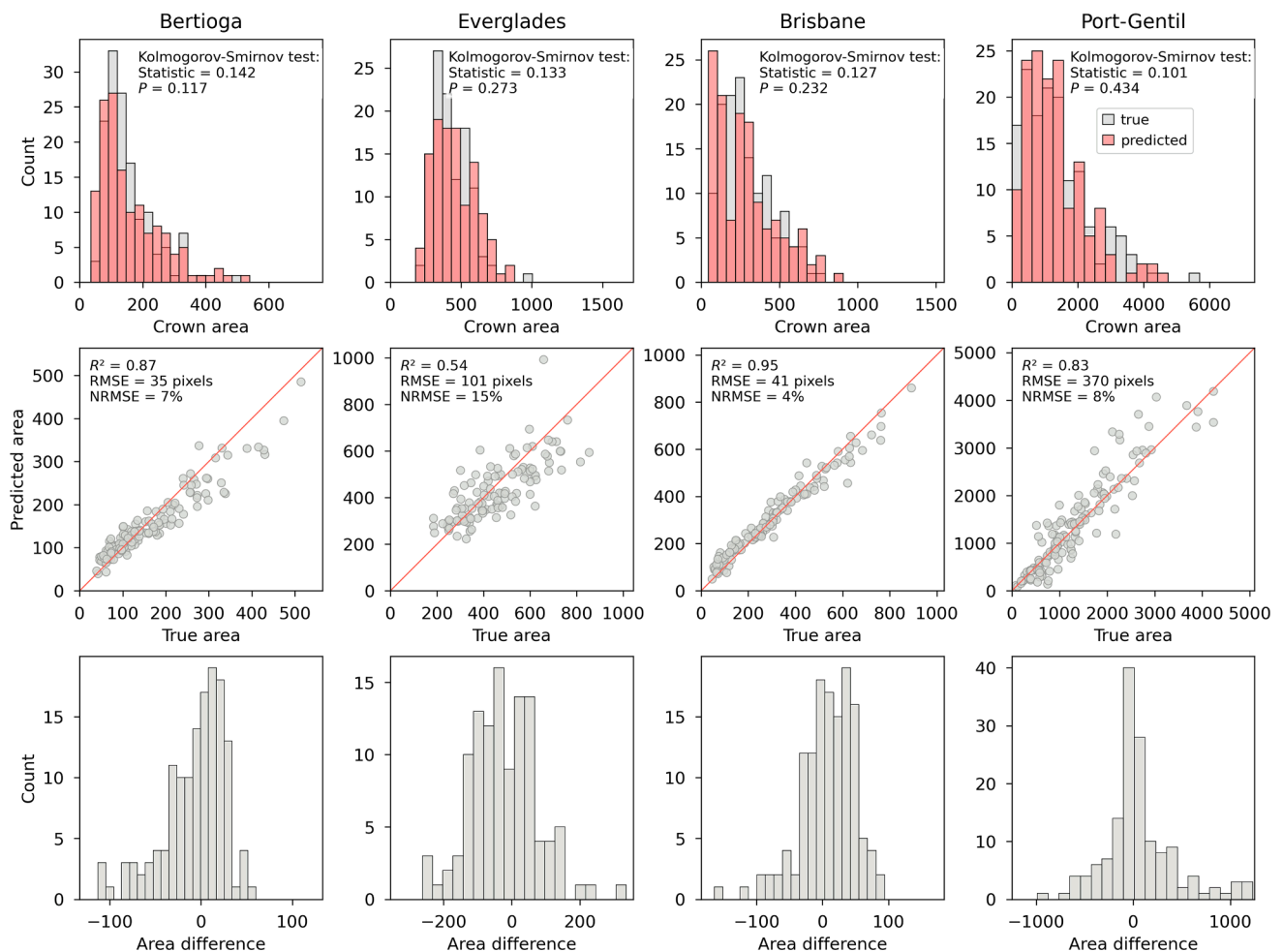


Fig. 7. (Upper row) Distribution of the true and predicted crown areas, (Middle row) true vs. predicted crown area, and (Lower row) distribution of the area difference (in pixel units) between the measured and predicted crowns for the four study sites using the PAN band only. Only the predicted crowns sharing an Intersection over Union (IoU) $\geq 50\%$ with the true crowns were considered here.

3.5. Comparison with other methods

The results of the ITC delineation obtained with the different methods are summarized in Table 2. Regarding Precision, Recall, and F1-score, the proposed method outperformed the previously published methods, namely the marker-controlled watershed, the region-growing, and the multiresolution segmentations, as well as the Mask R-CNN. Applying the marker-controlled watershed segmentation to the original PAN images led to the inaccurate delineation of ITCs, as indicated by average Precision and Recall values ≤ 0.66 , regardless of the site.

Watershed segmentation usually results in oversegmentation of the image, expressed here as fragmentation of a single true crown into several smaller crown polygons. Distinguishing mangrove crowns from the original PAN image is highly challenging, thus explaining the poor performances observed with this method. As indicated in Table 2, the proposed method enhanced the border of the crown, thus easing the delineation by the watershed segmentation algorithm as compared to applying the latter alone. The region-growing and -merging (multi-resolution) segmentations provided good performances, with F1-scores ≥ 0.70 in Bertioga, Brisbane, and Port-Gentil. Since these algorithms

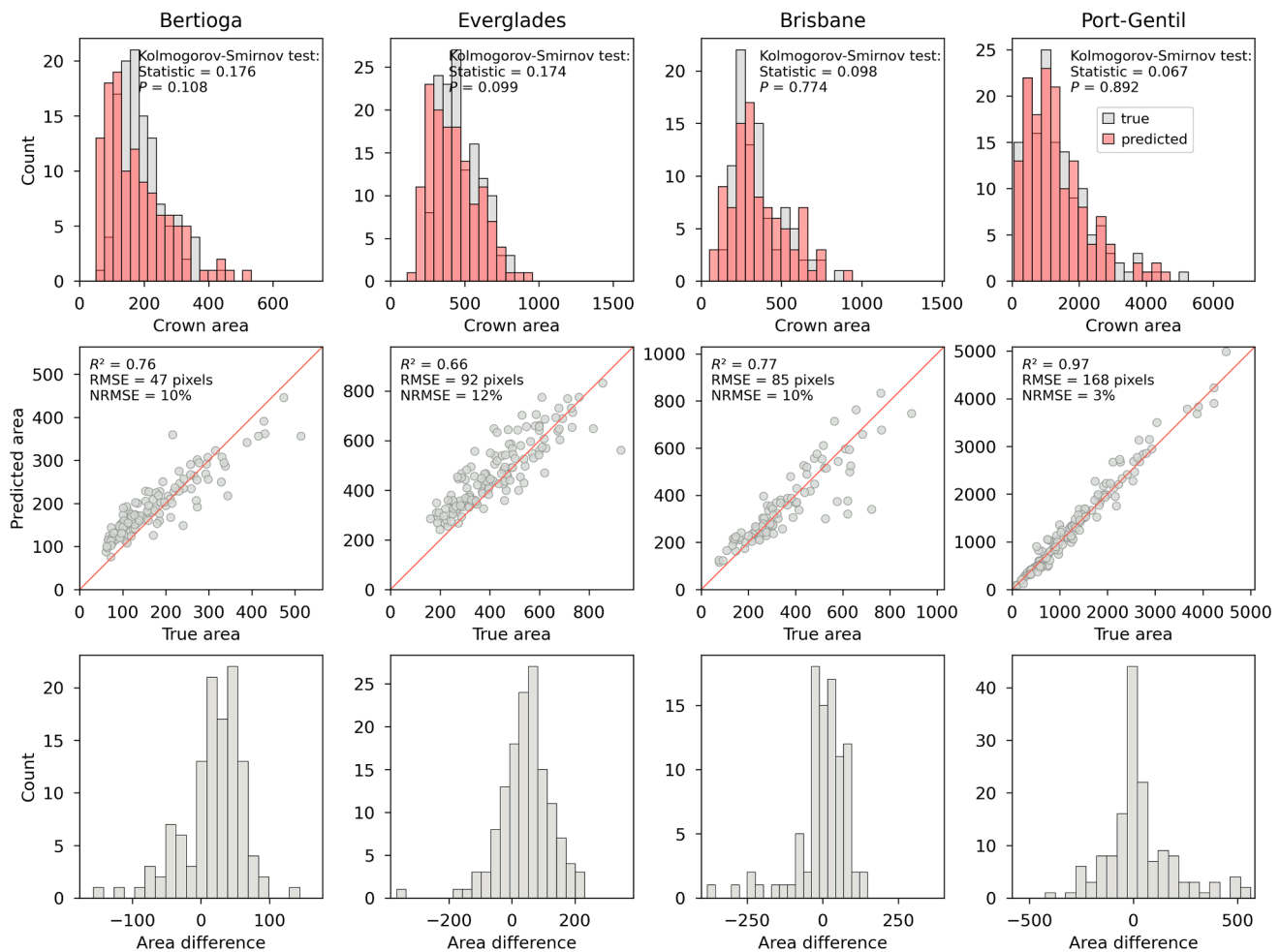


Fig. 8. (Upper row) Distribution of the true and predicted crown areas, (Middle row) true vs. predicted crown area, and (Lower row) distribution of the area difference (in pixel units) between the measured and predicted crowns for the four study sites using the VNIR bands (NIR-red-green). Only the predicted crowns sharing an Intersection over Union (IoU) $\geq 50\%$ with the true crowns were considered here. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

exploit more bands, they accurately distinguish treetops, crown, and non-crown pixels. However, they suffer from serious limitations when crown separability is limited; or when high variability in crown area is observed. Based on our observations, the region-growing algorithm is well-suited to delineate isolated trees but tends to produce large segments by merging clumped crowns and by considering pixels of understory vegetation in crown segments. Also, we noticed that the multiresolution segmentation ensures the homogeneity of segmented crowns – controlled by the shape and compactness parameters – but requires a fixed scale parameter as input. Consequently, this algorithm does not handle variability in the crown area, as observed in the Everglades. It results either in the undersegmentation of small crowns (high scale value) or in the oversegmentation of large crowns (low scale value). The same limitation was observed with the Mask R-CNN. Its performances outperformed previously-published methods but remained substantially worse than our MT-EDv3-based method, especially on sites with high variability of crown area. Thus, none of the previously published methods proved suitable to delineate mangrove ITCs accurately and across sites. Our method demonstrates that, by isolating crowns from the background and by enhancing their edges, ITC delineation can be achieved accurately in mangroves using satellite imagery, regardless of the tree's configuration (isolated, clumped, fringe) and crown area and shape.

4. Discussion

The RS community has well-regarded the delineation of ITCs based on satellite imagery in temperate, tropical, boreal, and urban forests (Aval et al., 2018; Dalponte et al., 2014; Tochon et al., 2015; Wagner et al., 2018). To address the backlog observed for mangroves, we proposed a method that partially oversteps the crown clumping and fragmentation problem while minimizing understory vegetation's influence on crown separability. The newly-developed method succeeded in delineating a wide range of tree crowns. In this section, we discuss the sources of inaccuracies and the possible improvements identified in this work, which should be addressed in further research.

4.1. Sources of inaccuracies

4.1.1. Viewing geometry

The inaccuracies observed in the delineation of mangrove ITCs can be attributed to two distinct factors: the viewing geometry of images and the structural heterogeneity of the forest. Crown delineation is indeed greatly influenced by the viewing geometry of images. In particular, off-nadir view improves the temporal resolution, but it thoroughly compromises treetop detection and crown delineation by masking crown borders and inter-crown spaces (Kempf et al., 2021). The geometric and radiometric consequences of far-nadir view are still poorly documented (Ke and Quackenbush, 2011a). Still, authors agree on the point that,

Table 2

Precision, Recall, and F1-score computed between the true and predicted ITCs for the four study sites, using three previously-proposed delineation methods, the Mask R-CNN, and the newly-developed method (see Section 2.6 for details about the methods). For each site, the mean ($\pm$ SD) is presented and the best result is marked bold.

	<i>Precision</i>	<i>Recall</i>	<i>F1-score</i>
<u>Bertioga:</u>			
Proposed method (PAN)	0.88 ($\pm$ 0.11)	0.85 ($\pm$0.14)	0.85 ($\pm$0.07)
Proposed method (VNIR)	0.93 ($\pm$0.10)	0.78 ($\pm$ 0.15)	0.83 ($\pm$ 0.08)
Marker-controlled watershed	0.51 ($\pm$ 0.06)	0.55 ($\pm$ 0.12)	0.53 ($\pm$ 0.09)
Region-growing (Ecognition®)	0.56 ($\pm$ 0.14)	0.73 ($\pm$ 0.06)	0.63 ($\pm$ 0.10)
Multiresolution segmentation (Ecognition®)	0.71 ($\pm$ 0.11)	0.77 ($\pm$ 0.1)	0.74 ($\pm$ 0.10)
Mask R-CNN	0.82 ($\pm$ 0.07)	0.78 ($\pm$ 0.12)	0.79 ($\pm$ 0.09)
<u>Everglades:</u>			
Proposed method (PAN)	0.78 ($\pm$ 0.11)	0.88 ($\pm$0.14)	0.81 ($\pm$ 0.07)
Proposed method (VNIR)	0.89 ($\pm$0.10)	0.82 ($\pm$ 0.13)	0.85 ($\pm$0.07)
Marker-controlled watershed	0.41 ($\pm$ 0.12)	0.49 ($\pm$ 0.15)	0.43 ($\pm$ 0.12)
Region-growing (Ecognition®)	0.62 ($\pm$ 0.04)	0.50 ($\pm$ 0.09)	0.55 ($\pm$ 0.06)
Multiresolution segmentation (Ecognition®)	0.62 ($\pm$ 0.11)	0.58 ($\pm$ 0.07)	0.60 ($\pm$ 0.08)
Mask R-CNN	0.66 ($\pm$ 0.08)	0.64 ($\pm$ 0.11)	0.65 ($\pm$ 0.10)
<u>Brisbane:</u>			
Proposed method (PAN)	0.93 ($\pm$0.07)	0.84 ($\pm$0.14)	0.87 ($\pm$0.08)
Proposed method (VNIR)	0.87 ($\pm$ 0.12)	0.81 ($\pm$ 0.14)	0.83 ($\pm$ 0.08)
Marker-controlled watershed	0.51 ($\pm$ 0.16)	0.58 ($\pm$ 0.60)	0.54 ($\pm$ 0.70)
Region-growing (Ecognition®)	0.66 ($\pm$ 0.05)	0.74 ($\pm$ 0.08)	0.70 ($\pm$ 0.07)
Multiresolution segmentation (Ecognition®)	0.71 ($\pm$ 0.14)	0.74 ($\pm$ 0.14)	0.72 ($\pm$ 0.09)
Mask R-CNN	0.79 ($\pm$ 0.05)	0.78 ($\pm$ 0.09)	0.78 ($\pm$ 0.08)
<u>Port-Gentil:</u>			
Proposed method (PAN)	0.84 ($\pm$ 0.12)	0.81 ($\pm$ 0.08)	0.83 ($\pm$ 0.09)
Proposed method (VNIR)	0.92 ($\pm$0.09)	0.95 ($\pm$0.07)	0.93 ($\pm$0.06)
Marker-controlled watershed	0.66 ($\pm$ 0.16)	0.61 ($\pm$ 0.08)	0.63 ($\pm$ 0.11)
Region-growing (Ecognition®)	0.70 ($\pm$ 0.09)	0.79 ($\pm$ 0.12)	0.74 ($\pm$ 0.08)
Multiresolution segmentation (Ecognition®)	0.76 ($\pm$ 0.13)	0.81 ($\pm$ 0.09)	0.78 ($\pm$ 0.11)
Mask R-CNN	0.80 ($\pm$ 0.11)	0.82 ($\pm$ 0.10)	0.81 ($\pm$ 0.11)

above a certain angle – which depends upon the sensor and forest characteristics, the delineation of ITCs becomes hugely challenging due to occlusions and cast shadows (Gougeon, 1999; Ke and Quackenbush, 2011b; Kempf et al., 2021). This limitation of satellite imagery has been commonly observed for various forest stands and spaceborne sensors such as WorldView, QuickBird, and IKONOS (Ferreira et al., 2021; Ke et al., 2010; Song et al., 2010). In our case, the variability in the viewing geometry among study sites may have contributed to better performance on sites with low off-nadir view. This is corroborated by a better delineation of ITCs from the Bertioga, Brisbane, and Port-Gentil images, (all lying between 11.5 and 13.4° off-nadir) than from the 17° off-nadir Everglades image. To quantitatively assess the influence of the off-nadir viewing on the resulting delineation, we conducted a sensitivity analysis using two additional WorldView-2 and –3 PAN images acquired over the same area in the Bertioga site (Fig. 9). While the 12.1° off-nadir image used in this study proved suited to accurately delineate ITCs (average F1-score = 0.85), applying the same method at 24.9° off-nadir drastically worsened the quality of the delineation (average F1-score = 0.71). At a higher angle (26.5°) and using a WorldView-2 image with coarser resolution (0.52 m), crowns become almost indistinguishable and are thus poorly delineated (average F1-score = 0.45). Although those images were acquired over 11 months, no substantial change in the true shape and size of crowns is likely to have occurred as a result of tree growth (De Alvarenga et al., 2017). Therefore, most of the differences observed among these images can be attributed to differences in the viewing geometry.

4.1.2. Forest structural heterogeneity

Crown delineation is greatly facilitated in homogeneous forest stands with low species richness and evenly aged and spaced trees such as tree plantations (Hornero et al., 2020; Johansen et al., 2020; Wagner et al., 2019). In mangroves, however, high within-stand heterogeneity is

observed due to species zonation, environmental gradients, understory vegetation establishment, and natural and anthropogenic perturbations (Amir and Duke, 2019; Kamal et al., 2015; Li et al., 2021; Vovides et al., 2018). Such heterogeneity results in crowns with varying size, shape, and degree of clumping or isolation, affecting the accuracy of the delineation.

Tree configuration, which refers to the location of a tree in the forest and its proximity to surrounding trees (e.g., clumped, isolated, fringe) may explain some inaccuracies in the delineation of mangrove ITCs observed here. For example, the Brisbane mangrove has a higher proportion of isolated trees located in the landward and seaward shorelines (see example in Fig. 10). Since those trees are isolated from the rest of the forest and surrounded by either bare sediments or water, their crown is detected in the PAN and the VNIR images, thus explaining the relatively low influence of band selection on the quality of the segmentation for this site.

In mangroves, tree clumping appears as an overlapping of the crown edges, making ITC delineation particularly difficult. Edge pixels appear usually less bright than treetop pixels in the panchromatic and VNIR images. Consequently, the MT-EDv3 network attributes low normalized Euclidean distance value to overlapping edge pixels, while high values are predicted for treetops pixels. Then, the LoG filter further enhances the separability of clumped crowns, allowing individual discrimination of partially overlapping ITCs, unlike previous methods. Crown clumping from distinct individuals is widely observed in the mangroves of Bertioga and the Everglades. This phenomenon results in narrowing the inter-crown spaces – sometimes one or two pixels wide in the PAN image, which can be lost in the pansharpening of 1.2-m VNIR images. Coupled with a large off-nadir angle, this may explain the lower accuracy observed in the Everglades compared to the other sites. Based on field observations, the clumping of multiple sub-crowns from a single individual – as defined by Clough et al. (1997) and illustrated in

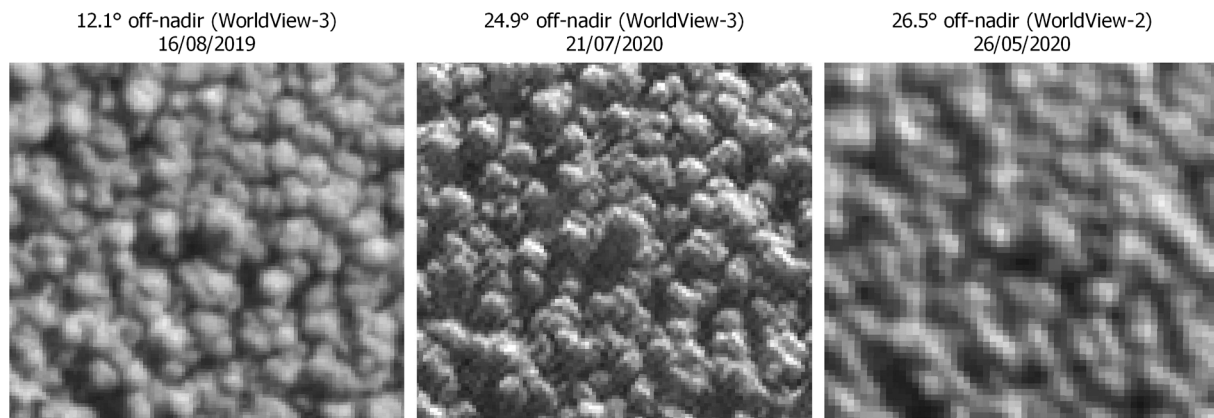


Fig. 9. Influence of the viewing geometry on the detectability and separability of mangrove ITCs. These images show the same subset of the Bertioga mangrove forest as seen from WorldView-2 and -3 PAN imagery with various off-nadir angles.

Supplementary Fig. S7 – also contributes to the spatial heterogeneity of mangrove forests. These sub-crowns are characterized by a narrow spacing from each other and by a larger spacing from surrounding tree crowns (Fig. 10). Using the appropriate patch size in the MT-EDv3 network, they appear as a unique crown in the distance image, avoiding oversegmentation. Conversely, such heterogeneity compromises the performance of other region-growing and -merging algorithms, limiting their applicability to mangrove forests.

Handling the variability in crown area and shape is a critical challenge in ITC delineation. Many methods require fixed parameters to control the segment area, compactness, or roundness (such as the multiresolution segmentation (Johansen et al., 2020; Kamal et al., 2016)), making it difficult to accurately delineate mangrove trees with irregularly-shaped and -sized crowns, as observed in our study sites. The same limitation applies to the Mask R-CNN, which is highly sensitive to variations in the size and shape of ITCs (Ferreira et al., 2020). Our method overcomes this issue by training the MT-EDv3 network on a set of representative crowns with varying characteristics, therefore considering the within-stand heterogeneity. Building a large dataset of training ITCs that reflects the forest heterogeneity is therefore critical to ensure an accurate automatic delineation. It is worth noting that manual delineation is prone to errors due to mangrove-specific characteristics (e.g. crown clumping) and image-related factors (e.g. viewing and illumination geometry). However, performing manual delineation by photo-interpretation is the only way of obtaining a large and robust dataset of ITCs ($n = 2000$ in our case) over multiple, distant sites with highly diverse crown structural properties. This time-consuming and rigor-demanding step is not necessary when using unsupervised methods such as the multiresolution segmentation, making them faster and easier to apply. However, as highlighted here, these methods are not reliable across sites, especially in forests with high heterogeneity. Apart from tree age and environmental factors, crown characteristics are determined genetically by the species so that high species richness results in more forest heterogeneity (Li et al., 2019). Although quantifying the influence of species richness on ITC delineation is out of the scope of this study, we bring evidence of possible links between species patterns in mangrove forests and the accuracy of crown delineation. In Bertioga or Brisbane, mangrove species form a distinct zonation, so some species are found mainly in the fringe or the landward parts of the forest (Flores de Santiago, 2013; Kamal et al., 2015). In this case, the spatial heterogeneity is reduced as species are organized in homogeneous units, so accurate crown delineation was achieved from the PAN with minor benefit of adding the VNIR bands. In the Everglades and Port-Gentil, however, mixed mangrove forests dominate, leading to a more complex forest structure (Ruiz et al., 2018), so using the VNIR images instead of the PAN benefited the delineation of neighboring crowns from different species, as the spectral information contained in the green, red, and NIR

band helped to better distinguish crown borders (see **Supplementary Fig. S8**).

As another contributor to forest heterogeneity, understory vegetation is one of the most troublesome factors in the delineation of crowns in many forest ecosystems (Campbell et al., 2020; Fassnacht et al., 2016; Huang et al., 2018; Wagner et al., 2018). Vegetation growing directly below the canopy has only little influence on the delineation as the upper crowns masks it. However, low-strata vegetation fills the spaces between crown borders, thus mimicking crown clumping and leading to undersegmentation. Understory vegetation may differ in species composition from the upper canopy, making it maskable from multispectral imagery. Such discrimination becomes extremely challenging when the understory vegetation is similar to that forming the upper canopy as, unlike active optical RS, passive multispectral imagery does not inform the height of the observed vegetation (Li et al., 2021; Taureau et al., 2019). This limitation of passive imagery caused inaccuracies in the delineation of ITCs in our case. To illustrate this, we surveyed several locations of the Bertioga mangrove with contrasted densities of understory vegetation (Fig. 11). In this forest, the understory vegetation is made of the same species as the upper canopy, but an important heterogeneity is observed within the site. Bare sediment areas (Fig. 11a) make inter-crown spaces appear as dark pixels, while sparse to dense mangrove saplings (Fig. 11b-c) result in green-dark mixed. Finally, most of the inaccuracies in crown delineation may be attributed to areas covered by young mangrove trees (2–3-m height, Fig. 11d) because of inter-crown space-filling. As this interpretation is based on field observations, further research should quantitatively address the influence of understory vegetation and its characteristics (e.g., species, height, density) on ITC delineation.

Hence, handling forest structural heterogeneity is a key challenge of crown delineation in mangrove forests. As emphasized by our study, crown enhancing can help to cope with the variability in crown shape and area while limiting the influence of understory vegetation. Since our method is not free from defects, it could be improved in multiple ways, for example, by coupling multispectral imagery to other RS data sources. Besides, the transferability of our method to other passive optical sensors should be evaluated towards operational applications by end-users. These points are discussed in the following sections.

4.2. Generalization of the method

Because mangrove forests flourish in tropical and neotropical areas, obtaining high-quality satellite images is challenging due to the high probability of cloud cover. Ensuring the transferability of methods to a wide range of satellite sensors is, therefore, crucial to maximizing image availability. Because it requires only the PAN channel or a few VNIR bands, our method can be applied to other satellite sensors if they

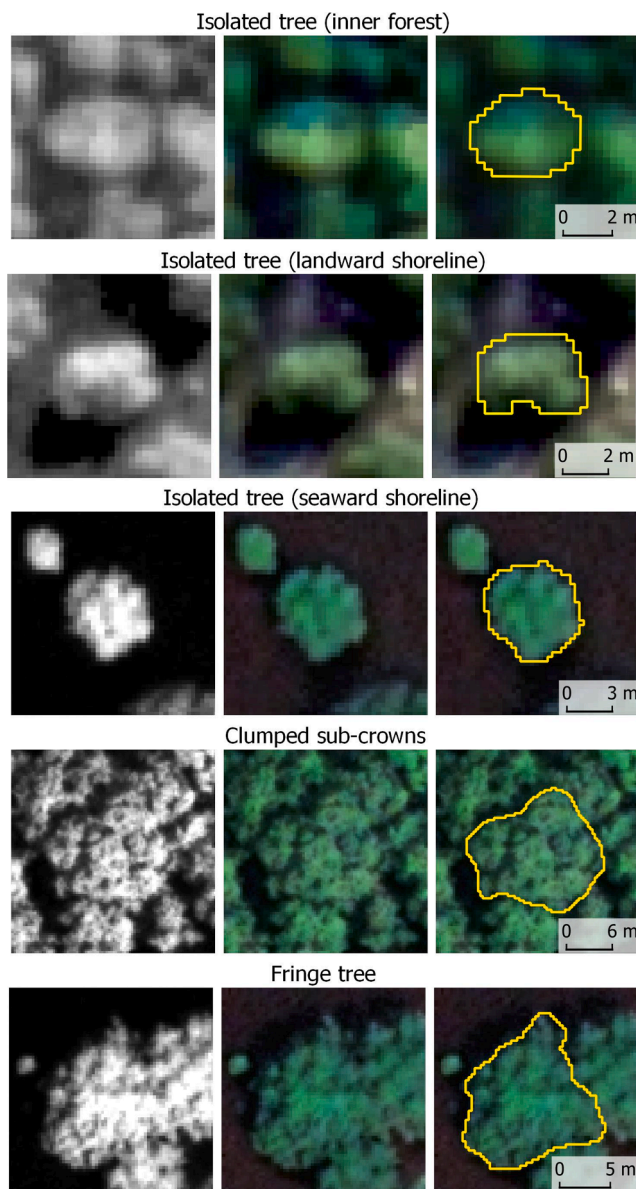


Fig. 10. Examples of successful individual tree crown delineation using the proposed method for various configurations of mangrove trees. From left to right: PAN view, RGB view, and RGB view + automatically delineated crown polygon. For graphical convenience, the surrounding delineated crown polygons were removed.

comply with the recommended spatial resolution (pixel size ≤ 1 m). As a general rule, the authors state that the spatial resolution must be smaller than the crown size to achieve accurate ITC delineation (Ke and Quackenbush, 2011a). Satellite sensors like IKONOS, QuickBird (archives), GeoEye-1, and Pleiades provide panchromatic images with a 0.46 to 0.82 m spatial resolution and already proved relevant to delineate ITCs in other forest ecosystems (Greenberg et al., 2005; Pu and Landry, 2020; Song et al., 2010), so they could be well-suited to do the same in mangrove forests using our method. Other RS platforms should be considered as well. Historically, aerial imagery has been widely used to delineate tree crowns (Ke and Quackenbush, 2011a). In that sense, our method can be easily transferred to panchromatic or multispectral aerial imagery with very high spatial resolution (pixel size < 1 m). Likewise, UAV-borne PAN and multispectral cameras show great potential as they make it possible to survey specific areas at the appropriate time with up to centimeter-scale resolution (Lassalle, 2021; Torresan et al., 2017). These cameras are now widely commercialized and plenty

of them cover the spectral bands recommended for applying our method. Several authors favored the use of UAV imagery over satellites in mangroves (Ruwaimana et al., 2018), but UAVs appear unsuited for broad-scale mapping, as a single mangrove stand can easily cover a hundred hectares (see Section 2.1). So, instead of an alternative, they should be seen as a complement to satellite imagery, as they could be used, for example, to manually outline ITCs for satellite applications and to help distinguish crown from non-crown vegetation. However, this would imply accurate registration between UAV and satellite images as done in other ecosystems (Johansen et al., 2020; Miranda et al., 2020).

Adding spectral information may also benefit the delineation of mangrove ITCs, as observed in our study (see results for PAN and VNIR in Sections 3). First, by introducing Short-Wave Infrared (SWIR) bands, which recently proved relevant to enhance mangrove pixels identification (Baloloy et al., 2020; Jia et al., 2019). Then, by increasing the spectral resolution of images. ITC delineation may benefit from hyperspectral imagery by refining the optimal spectral bands exploited in the method, as some bands may help to overcome the limitations mentioned in Section 4.1. However, the current and upcoming hyperspectral satellite sensors are limited to 8 to 30-m spatial resolution, so no applications are yet conceivable. Instead, airborne, and UAV-based hyperspectral imagery satisfies both high spatial and spectral resolution and should be considered in the method improvement.

VHR satellite imagery is not intended to address all the challenges in mangrove ITC delineation. As mentioned in Section 4.1, some limitations cannot be overcome by using solely passive optical RS. That is, delineating very small crowns (i.e. lower in width than the pixel size) remains highly difficult even if the optimal conditions are met (low off-nadir angle, low heterogeneity, and no understory vegetation). Coupling satellite imagery to CHMs obtained from UAV- or airborne-based LiDAR data with high point density could significantly improve the delineation accuracy. Heenkenda et al. (2015), found that combining a CHM to WorldView-2 imagery only slightly improves mangrove ITC delineation (+3% of OA). Conversely, UAV-based LiDAR data provide fine details about the canopy height and structure and achieve good delineation when exploited alone (Yin and Wang, 2019). Thus, LiDAR-derived CHM obtained from UAVs could be added as a layer to the MT-EDv3 network in our method, as recently attempted for mangrove biomass estimation (Qiu et al., 2019). This implies, however, a complex registration of images, and restricts the area coverage to that of UAV images. Hence, several ways are identified to improve the delineation of mangrove ITCs based on our method and should be addressed in future studies. In its current version, our method is nonetheless reliable and opens encouraging perspectives for monitoring mangrove forests.

4.3. Perspectives for mangrove monitoring

Trees are the angular stone of mangrove ecosystems. They are, however, vulnerable to natural and anthropogenic stressors and disturbances (Duke et al., 2007; Spalding and Leal, 2021). Satellite-based monitoring is an essential component of mangrove conservation since mangrove forests cover large areas with limited field access. In this respect, the need for ITC delineation is justified by many applications. First, it has a crucial role in assessing the consequences of disturbances on mangroves as it involves mapping tree dieback over large areas. For example, lightning strikes and oil spills cause the apparition of large gaps in mangrove canopies, characterized by bare areas surrounded by healthy trees (Amir and Duke, 2019) (see examples in Supplementary Fig. S9). These gaps go through several stages toward canopy closure, all varying in crown density and area. Although mangrove gaps can be detected accurately using LiDAR-derived CHMs, it remains unsuited to monitor their recovery (Zhang, 2008). Thanks to very-high spatial resolution, the method presented in our study could be applied to detect recent gaps by isolating non-crown areas and to evaluate the late recovery stages of gaps by estimating changes in the crown area within gaps over years.



Fig. 11. Spatial variability in understory vegetation observed in the mangrove forest of Bertioga, Brazil. (a) Bare sediments, (b) sparse and (c) dense mangrove saplings, and (d) young mangrove trees (2–3-m height).

Crown delineation is also the first, key step in object-based classification (e.g., species, health) and biophysical-biochemical parameter retrieval in satellite-based studies (Campbell et al., 2020; Fang et al., 2018; Ferreira et al., 2019; Huo et al., 2021; Schäfer et al., 2016; Wagner et al., 2018). Until now, applications in mangroves remained limited to pixel-based or inaccurate object-based mappings (Cao et al., 2018; Kamal et al., 2016). In particular, Wang et al. (2019b) pointed out the lack of studies on mangrove stress assessment, which might be related to the lack of accurate, broad-scale methods to delineate ITCs. Thanks to our method, accurate delineation can be obtained over large mangrove stands, paving the way to numerous ITC-based applications. CNNs like the MT-EDv3 network allow performing multiple tasks at the same time, including, for example, crown delineation and species classification or tree height estimation (Ferreira et al., 2021; Hao et al., 2021; Martins et al., 2021). Such applications could improve our method, allowing to perform species mapping, canopy biochemistry retrieval, blue carbon stock estimation, and stress assessment at a broad-scale towards improving the monitoring of mangrove forests and meeting the conservation goals of these invaluable ecosystems.

5. Conclusion

This study is the first to propose a practical method to delineate ITCs in mangrove forests using VHR satellite imagery. The method enhances crown separability using a deep learning model and a LoG filter to

facilitate the automatic delineation of crowns by a marker-controlled watershed segmentation. The proposed method succeeded in delineating ITCs over four archetype mangrove forests with varying characteristics while limiting the influence of confounding artifacts such as understory vegetation and crown clumping. It outperformed previously published methods based on satellite imagery and another widely-used deep learning segmentation framework. Our method is well-adapted to WorldView imagery and shows promising potential for application to other VHR satellite sensors and aerial and UAV imagery. Increasing attention is given to mangrove ecosystems, motivating the development of new RS-based solutions to monitor these ecosystems. Since ITC delineation is critical for mangrove mapping, our method could serve as a basis for many applications, from tree counting to health assessment.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was performed in the frame of the Active and Passive Imagery for Oil Monitoring in Mangroves and Marine Environments (APIOM³E) project between the University of Campinas and

TotalEnergies Brazil through the Agência Nacional do Petróleo, Gás Natural e Biocombustíveis (ANP) project n° SAP 4600002921 (22-P-05553/2020 - 92105). It benefited from the Fundação de Desenvolvimento da Unicamp (FUNCAMP) grant n°5596 to G. Lassalle. The authors gratefully acknowledge Anthony Credo, Dominique Dubucq, and Dimas Coelho for their implications in the project, and MAXAR for providing the WorldView imagery used in this study. C.R Souza Filho, M. P. Ferreira, and L.E.C. La Rosa were supported by the Brazilian National Council for Scientific and Technological Development (CNPq) grants #309712/2017-30, #306345/2020-0 and #141205/2018-0.

Declarations of interest: None.

Funding

Financial support for this work was provided by TotalEnergies.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.isprsjprs.2022.05.002>.

References

- Aldous, A., Schill, S., Raber, G., Paiz, M.-C., Mambela, E., Stévar, T., Horning, N., Friess, D., 2021. Mapping complex coastal wetland mosaics in Gabon for informed ecosystem management: use of object-based classification. *Remote Sens. Ecol. Conserv.* 7 (1), 64–79. <https://doi.org/10.1002/rse2.161>.
- Alongi, D.M., 2012. Carbon sequestration in mangrove forests. *Carbon Manage.* 3 (3), 313–322. <https://doi.org/10.4155/cmt.12.20>.
- Amir, A.A., Duke, N.C., 2019. Distinct characteristics of canopy gaps in the subtropical mangroves of Moreton Bay, Australia. *Estuar. Coast. Shelf Sci.* 222, 66–80. <https://doi.org/10.1016/j.ecss.2019.04.007>.
- Aval, J., Demuyneck, J., Zenou, E., Fabre, S., Sheeren, D., Fauvel, M., Adeline, K., Briottet, X., 2018. Detection of individual trees in urban alignment from airborne data and contextual information: A marked point process approach. *ISPRS J. Photogramm. Remote Sens.* 146, 197–210. <https://doi.org/10.1016/j.isprsjprs.2018.09.016>.
- Baloloy, A.B., Blanco, A.C., Sta. Ana, R.R.C., Nadaoka, K., 2020. Development and application of a new mangrove vegetation index (MVI) for rapid and accurate mangrove mapping. *ISPRS J. Photogramm. Remote Sens.* 166, 95–117. <https://doi.org/10.1016/j.isprsjprs.2020.06.001>.
- Barbier, E.B., 2016. The protective service of mangrove ecosystems: A review of valuation methods. *Mar. Pollut. Bull.* 109 (2), 676–681. <https://doi.org/10.1016/j.marpolbul.2016.01.033>.
- Biswas, H., Zhang, K., Ross, M.S., Gann, D., 2020. Delineation of Tree Patches in a Mangrove-Marsh Transition Zone by Watershed Segmentation of Aerial Photographs. *Remote Sensing* 12, 2086. <https://doi.org/10.3390/rs12132086>.
- Braga, J.R.R., Peripato, V., Dalagnol, R., Ferreira, M.P., Tarabalka, Y., Aragão, L.E.O.C., de Campos Velho, H.F., Shigemori, E.H., Wagner, F.H., 2020. Tree Crown Delineation Algorithm Based on a Convolutional Neural Network. *Remote Sensing* 12, 1288. <https://doi.org/10.3390/rs12081288>.
- Bunting, P., Lucas, R., 2006. The delineation of tree crowns in Australian mixed species forests using hyperspectral Compact Airborne Spectrographic Imager (CASI) data. *Remote Sens. Environ.* 101 (2), 230–248. <https://doi.org/10.1016/j.rse.2005.12.015>.
- Bunting, P., Rosenqvist, A., Lucas, R.M., Rebelo, L.-M.-M., Hilarides, L., Thomas, N., Hardy, A., Itoh, T., Shimada, M., Finlayson, C.M., 2018. The global mangrove watch - A new 2010 global baseline of mangrove extent. *Remote Sensing* 10, 1669. <https://doi.org/10.3390/rs10101669>.
- Byrd, K.B., Ballanti, L., Thomas, N., Nguyen, D., Holmquist, J.R., Simard, M., Windham-Myers, L., 2018. A remote sensing-based model of tidal marsh aboveground carbon stocks for the conterminous United States. *ISPRS J. Photogramm. Remote Sens.* 139, 255–271. <https://doi.org/10.1016/j.isprsjprs.2018.03.019>.
- Campbell, M.J., Dennison, P.E., Tune, J.W., Kannenberg, S.A., Kerr, K.L., Coddling, B.F., Anderregg, W.R.L., 2020. A multi-sensor, multi-scale approach to mapping tree mortality in woodland ecosystems. *Remote Sens. Environ.* 245, 111853. <https://doi.org/10.1016/j.rse.2020.111853>.
- Cao, J., Leng, W., Liu, K., Liu, L., He, Z., Zhu, Y., 2018. Object-Based Mangrove Species Classification Using Unmanned Aerial Vehicle Hyperspectral Images and Digital Surface Models. *Remote Sensing* 10, 89. <https://doi.org/10.3390/rs10010089>.
- Castaneda-Moya, E., Twilley, R.R., Rivera-Monroy, V.H., Marx, B.D., Coronado-Molina, C., Ewe, S.M.L., 2011. Patterns of Root Dynamics in Mangrove Forests Along Environmental Gradients in the Florida Coastal Everglades, USA. *Ecosystems* 14 (7), 1178–1195. <https://doi.org/10.1007/s10021-011-9473-3>.
- Chemura, A., van Duren, I., van Leeuwen, L.M., 2015. Determination of the age of oil palm from crown projection area detected from WorldView-2 multispectral remote sensing data: The case of Ejisu-Juaben district, Ghana. *ISPRS J. Photogramm. Remote Sens.* 100, 118–127. <https://doi.org/10.1016/j.isprsjprs.2014.07.013>.
- Chen, L.-C., Zhu, Y., Papandreou, G., Schroff, F., Adam, H., 2018. Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation. pp. 833–851. https://doi.org/10.1007/978-3-030-01234-2_49.
- Chiang, C.-Y., Barnes, C., Angelov, P., Jiang, R., 2020. Deep Learning-Based Automated Forest Health Diagnosis From Aerial Images. *IEEE Access* 8, 144064–144076. <https://doi.org/10.1109/ACCESS.2020.3012417>.
- Clough, B.F., Dixon, P., Dalhaus, O., 1997. Allometric Relationships for Estimating Biomass in Multi-stemmed Mangrove Trees. *Aust. J. Bot.* 45 (6), 1023. <https://doi.org/10.1071/BT96075>.
- Cohen, J., 1968. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychol. Bull.* 70, 213–220. <https://doi.org/10.1037/h0026256>.
- Dalpoite, M., Ørka, H.O., Ene, L.T., Gobakken, T., Næsset, E., 2014. Tree crown delineation and tree species classification in boreal forests using hyperspectral and ALS data. *Remote Sens. Environ.* 140, 306–317. <https://doi.org/10.1016/j.rse.2013.09.006>.
- De Alvarenga, A.M.S.B., Botosso, P.C., Soffiatti, P., 2017. Stem growth and phenology of three subtropical mangrove tree species. *Brazilian Journal of Botany* 40 (4), 907–914. <https://doi.org/10.1007/s40415-017-0397-9>.
- Duke, N.C., 2016. Oil spill impacts on mangroves: Recommendations for operational planning and action based on a global review. *Mar. Pollut. Bull.* 109 (2), 700–715. <https://doi.org/10.1016/j.marpolbul.2016.06.082>.
- Duke, N.C., Meynecke, J.-O., Dittmann, S., Ellison, A.M., Anger, K., Berger, U., Cannicci, S., Diele, K., Ewel, K.C., Field, C.D., Koedam, N., Lee, S.Y., Marchand, C., Nordhaus, I., Dahdouh-Guebas, F., 2007. A world without mangroves? *Science* 317 (5834), 41–42.
- Fang, F., McNeil, B.E., Warner, T.A., Maxwell, A.E., 2018. Combining high spatial resolution multi-temporal satellite data with leaf-on LiDAR to enhance tree species discrimination at the crown level. *Int. J. Remote Sens.* 39 (23), 9054–9072. <https://doi.org/10.1080/01431161.2018.1504343>.
- Fassnacht, F.E., Latifi, H., Stereńczak, K., Modzelewska, A., Lefsky, M., Waser, L.T., Straub, C., Ghosh, A., 2016. Review of studies on tree species classification from remotely sensed data. *Remote Sens. Environ.* 186, 64–87. <https://doi.org/10.1016/j.rse.2016.08.013>.
- Ferreira, M.P., Almeida, D.R.A.d., Papa, D.d.A., Minervino, J.B.S., Veras, H.F.P., Formighieri, A., Santos, C.A.N., Ferreira, M.A.D., Figueiredo, E.O., Ferreira, E.J.L., 2020. Individual tree detection and species classification of Amazonian palms using UAV images and deep learning. *For. Ecol. Manage.* 475, 118397. <https://doi.org/10.1016/j.foreco.2020.118397>.
- Ferreira, M.P., Féret, J.-B., Grau, E., Gastellu-Etchegorry, J.-P., do Amaral, C.H., Shimabukuro, Y.E., de Souza Filho, C.R., 2018. Retrieving structural and chemical properties of individual tree crowns in a highly diverse tropical forest with 3D radiative transfer modeling and imaging spectroscopy. *Remote Sens. Environ.* 211, 276–291. <https://doi.org/10.1016/j.rse.2018.04.023>.
- Ferreira, M.P., Lotte, R.G., D'Elia, F.V., Stamatopoulos, C., Kim, D.-H., Benjamin, A.R., 2021. Accurate mapping of Brazil nut trees (Bertholletia excelsa) in Amazonian forests using WorldView-3 satellite images and convolutional neural networks. *Ecol. Inf.* 63, 101302. <https://doi.org/10.1016/j.ecoinf.2021.101302>.
- Ferreira, M.P., Wagner, F.H., Aragão, L.E.O.C., Shimabukuro, Y.E., de Souza Filho, C.R., 2019. Tree species classification in tropical forests using visible to shortwave infrared WorldView-3 images and texture analysis. *ISPRS J. Photogramm. Remote Sens.* 149, 119–131. <https://doi.org/10.1016/j.isprsjprs.2019.01.019>.
- Flores de Santiago, F.J., 2013. Multiple approaches for assessing mangrove biophysical and biochemical variables using in situ and remote sensing techniques. PhD thesis dissertation from the The School of Graduate and Postdoctoral Studies. The University of Western Ontario, Ontario, Canada.
- Gougeon, F., 1999. Automatic individual tree crown delineation using a valley-following algorithm and a rule-based system, International Forum on Automated Interpretation of High Spatial Resolution Digital Imagery. *Forestry* 11–24.
- Greenberg, J.A., Dobrowski, S.Z., Ustin, S.L., 2005. Shadow allometry: Estimating tree structural parameters using hyperspatial image analysis. *Remote Sens. Environ.* 97 (1), 15–25. <https://doi.org/10.1016/j.rse.2005.02.015>.
- Hao, Z., Lin, L., Post, C.J., Mikhailova, E.A., Li, M., Chen, Y., Yu, K., Liu, J., 2021. Automated tree-crown and height detection in a young forest plantation using mask region-based convolutional neural network (Mask R-CNN). *ISPRS J. Photogramm. Remote Sens.* 178, 112–123. <https://doi.org/10.1016/j.isprsjprs.2021.06.003>.
- Hati, J.P., Goswami, S., Samanta, S., Pramanick, N., Majumdar, S.D., Chaube, N.R., Misra, A., Hazra, S., 2020. Estimation of vegetation stress in the mangrove forest using AVIRIS-NG airborne hyperspectral data. *Modeling Earth Systems and Environment* 7 (3), 1877–1889. <https://doi.org/10.1007/s40808-020-00916-5>.
- He, K., Gkioxari, G., Dollár, P., Girshick, R., 2017. Mask R-CNN, in: 2017 IEEE International Conference on Computer Vision (ICCV). IEEE, pp. 2980–2988. <https://doi.org/10.1109/ICCV.2017.322>.
- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep Residual Learning for Image Recognition. In: in: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, pp. 770–778. <https://doi.org/10.1109/CVPR.2016.90>.
- Heenkenda, M.K., Joyce, K.E., Maier, S.W., 2015. Mangrove Tree Crown Delineation from High-Resolution Imagery. *Photogramm. Eng. Remote Sens.* 81, 471–479. <https://doi.org/10.14358/PERS.81.6.471>.
- Heumann, B.W., 2011. Satellite remote sensing of mangrove forests: Recent advances and future opportunities. *Progress in Physical Geography: Earth and Environment* 35 (1), 87–108. <https://doi.org/10.1177/0309133310385371>.
- Hirata, Y., Tabuchi, R., Patanaponpaiboon, P., Pongparn, S., Yoneda, R., Fujioka, Y., 2014. Estimation of aboveground biomass in mangrove forests using high-resolution satellite data. *Journal of Forest Research* 19 (1), 34–41. <https://doi.org/10.1007/s10310-013-0402-5>.

- Hogarth, P.J., 2015. *The Biology of Mangroves and Seagrasses*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780198716549.001.0001>.
- Hornero, A., Hernández-Clemente, R., North, P.R.J., Beck, P.S.A., Boscia, D., Navas-Cortes, J.A., Zarco-Tejada, P.J., 2020. Monitoring the incidence of Xylella fastidiosa infection in olive orchards using ground-based evaluations, airborne imaging spectroscopy and Sentinel-2 time series through 3-D radiative transfer modelling. *Remote Sens. Environ.* 236, 111480. <https://doi.org/10.1016/j.rse.2019.111480>.
- Huang, H., Li, X.u., Chen, C., 2018. Individual Tree Crown Detection and Delineation From Very-High-Resolution UAV Images Based on Bias Field and Marker-Controlled Watershed Segmentation Algorithms. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 11 (7), 2253–2262. <https://doi.org/10.1109/JSTARS.2018.2830410>.
- Huo, L., Persson, H.J., Lindberg, E., 2021. Early detection of forest stress from European spruce bark beetle attack, and a new vegetation index: Normalized distance red & SWIR (NDRS). *Remote Sens. Environ.* 255, 112240. <https://doi.org/10.1016/j.rse.2020.112240>.
- Jia, Wang, Wang, Mao, Zhang, 2019. A New Vegetation Index to Detect Periodically Submerged Mangrove Forest Using Single-Tide Sentinel-2 Imagery. *Remote Sensing* 11, 2043. <https://doi.org/10.3390/rs11172043>.
- Johansen, K., Duan, Q., Tu, Y.-H., Searle, C., Wu, D., Phinn, S., Robson, A., McCabe, M.F., 2020. Mapping the condition of macadamia tree crops using multi-spectral UAV and WorldView-3 imagery. *ISPRS J. Photogramm. Remote Sens.* 165, 28–40. <https://doi.org/10.1016/j.isprsjprs.2020.04.017>.
- Kamal, M., Phinn, S., Johansen, K., 2016. Assessment of multi-resolution image data for mangrove leaf area index mapping. *Remote Sens. Environ.* 176, 242–254. <https://doi.org/10.1016/j.rse.2016.02.013>.
- Kamal, M., Phinn, S., Johansen, K., 2015. Object-Based Approach for Multi-Scale Mangrove Composition Mapping Using Multi-Resolution Image Datasets. *Remote Sensing* 7, 4753–4783. <https://doi.org/10.3390/rs70404753>.
- Kamal, M., Phinn, S., Johansen, K., 2014. Characterizing the Spatial Structure of Mangrove Features for Optimizing Image-Based Mangrove Mapping. *Remote Sensing* 6, 984–1006. <https://doi.org/10.3390/rs6020984>.
- Ke, Y., Quackenbush, L.J., 2011a. A review of methods for automatic individual tree-crown detection and delineation from passive remote sensing. *Int. J. Remote Sens.* 32 (17), 4725–4747. <https://doi.org/10.1080/01431161.2010.494184>.
- Ke, Y., Quackenbush, L.J., 2011b. A comparison of three methods for automatic tree crown detection and delineation from high spatial resolution imagery. *Int. J. Remote Sens.* 32 (13), 3625–3647. <https://doi.org/10.1080/01431161003762355>.
- Ke, Y., Zhang, W., Quackenbush, L.J., 2010. Active Contour and Hill Climbing for Tree Crown Detection and Delineation. *Photogramm. Eng. Remote Sens.* 76, 1169–1181. <https://doi.org/10.14358/PERS.76.10.1169>.
- Kempf, C., Tian, J., Kurz, F., D'Angelo, P., Schneider, T., Reinartz, P., 2021. Oblique view individual tree crown delineation. *Int. J. Appl. Earth Obs. Geoinf.* 99, 102314. <https://doi.org/10.1016/j.jag.2021.102314>.
- Krauss, K.W., Lovelock, C.E., McKee, K.L., López-Hoffman, L., Ewe, S.M.L., Sousa, W.P., 2008. Environmental drivers in mangrove establishment and early development: A review. *Aquat. Bot.* 89 (2), 105–127. <https://doi.org/10.1016/j.aquabot.2007.12.014>.
- Krauss, K.W., McKee, K.L., Lovelock, C.E., Cahoon, D.R., Saintilan, N., Reef, R., Chen, L., 2014. How mangrove forests adjust to rising sea level. *New Phytol.* 202 (1), 19–34. <https://doi.org/10.1111/nph.12605>.
- Kumar, T., Preeti, K., Chandrasekar, K., Bandyopadhyay, S., 2020. AVIRIS-NG Hyperspectral data for mapping mangrove forests and their health spatially: A case study of Indian Sundarbans. *J. Trop. For. Sci.* 32 (3), 317–331.
- La Rosa, L.E.C., Sothe, C., Feitosa, R.Q., de Almeida, C.M., Schimanski, M.B., Oliveira, D. A.B., 2021. Multi-task fully convolutional network for tree species mapping in dense forests using small training hyperspectral data. *ISPRS J. Photogramm. Remote Sens.* 179, 35–49. <https://doi.org/10.1016/j.isprsjprs.2021.07.001>.
- Lagomasino, D., Price, R.M., Whitman, D., Campbell, P.K.E., Melesse, A., 2014. Estimating major ion and nutrient concentrations in mangrove estuaries in Everglades National Park using leaf and satellite reflectance. *Remote Sens. Environ.* 154, 202–218. <https://doi.org/10.1016/j.rse.2014.08.022>.
- Lamparelli, C.C., Rodrigues, F.O., Moura, D.O., 1997. Long-term assessment of an oil spill in a mangrove forest in São Paulo, Brazil., in: UNESCO (Ed.), *Mangrove Ecosystem Studies in Latin America and Africa*. Paris, pp. 191–203.
- Lassalle, G., 2021. Monitoring natural and anthropogenic plant stressors by hyperspectral remote sensing: Recommendations and guidelines based on a meta-review. *Sci. Total Environ.* 788, 147758. <https://doi.org/10.1016/j.scitotenv.2021.147758>.
- Lewis, M., Pryor, R., Wilking, L., 2011. Fate and effects of anthropogenic chemicals in mangrove ecosystems: A review. *Environ. Pollut.* 159 (10), 2328–2346. <https://doi.org/10.1016/j.envpol.2011.04.027>.
- Li, Q., Wong, F.K.K., Fung, T., 2021. Mapping multi-layered mangroves from multispectral, hyperspectral, and LiDAR data. *Remote Sens. Environ.* 258, 112403. <https://doi.org/10.1016/j.rse.2021.112403>.
- Li, Q., Wong, F.K.K., Fung, T., 2019. Classification of Mangrove Species Using Combined WordView-3 and LiDAR Data in Mai Po Nature Reserve. *Hong Kong. Remote Sensing* 11, 2114. <https://doi.org/10.3390/rs11182114>.
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollar, P., 2017. Focal Loss for Dense Object Detection. In: in: 2017 IEEE International Conference on Computer Vision (ICCV). IEEE, pp. 2999–3007. <https://doi.org/10.1109/ICCV.2017.324>.
- Makowski, C., Finkl, C.W., 2018. Threats to Mangrove Forests. pp. E1–E1. https://doi.org/10.1007/978-3-319-73016-5_32.
- Manna, S., Raychaudhuri, B., 2020. Mapping distribution of Sundarban mangroves using Sentinel-2 data and new spectral metric for detecting their health condition. *Geocarto International* 35 (4), 434–452. <https://doi.org/10.1080/10106049.2018.1520923>.
- Marr, D., Hildreth, E., 1980. Theory of edge detection. *Proc. R. Soc. Lond. B Biol. Sci.* 207, 187–217. <https://doi.org/10.1098/rspb.1980.0020>.
- Martins, G.B., La Rosa, L.E.C., Happ, P.N., Filho, L.C.T.C., Santos, C.J.F., Feitosa, R.Q., Ferreira, M.P., 2021. Deep learning-based tree species mapping in a highly diverse tropical urban setting. *Urban For. Urban Greening* 64, 127241. <https://doi.org/10.1016/j.ufug.2021.127241>.
- Miranda, V., Pina, P., Heleno, S., Vieira, G., Mora, C., E.G.R. Schaefer, C., 2020. Monitoring recent changes of vegetation in Fildes Peninsula (King George Island, Antarctica) through satellite imagery guided by UAV surveys. *Science of the Total Environment* 704, 135295. <https://doi.org/10.1016/j.scitotenv.2019.135295>.
- Murphy, K.P., 2012. *Machine learning A probabilistic perspective*, The MIT Pr. ed. Cambridge, Massachusetts; London, England.
- Pastor-Guzman, J., Dash, J., Atkinson, P.M., 2018. Remote sensing of mangrove forest phenology and its environmental drivers. *Remote Sens. Environ.* 205, 71–84. <https://doi.org/10.1016/j.rse.2017.11.009>.
- Pouliot, D.A., King, D.J., Bell, F.W., Pitt, D.G., 2002. Automated tree crown detection and delineation in high-resolution digital camera imagery of coniferous forest regeneration. *Remote Sens. Environ.* 82 (2–3), 322–334. [https://doi.org/10.1016/S0034-4257\(02\)00050-0](https://doi.org/10.1016/S0034-4257(02)00050-0).
- Pu, R., Landry, S., 2020. Mapping urban tree species by integrating multi-seasonal high resolution pléiades satellite imagery with airborne LiDAR data. *Urban For. Urban Greening* 53, 126675. <https://doi.org/10.1016/j.ufug.2020.126675>.
- Qiu, Wang, Zou, Yang, Xie, Xu, Zhong, 2019. Finer Resolution Estimation and Mapping of Mangrove Biomass Using UAV LiDAR and WorldView-2 Data. *Forests* 10 (10), 871. <https://doi.org/10.3390/f10100871>.
- Ruiz, P.L., Prats, M.C., Perry, C.P., Garcia, A.A., Foguer, M.A., Ingram, J., Shamblin, R.B., Guichardot, M., Whelan, K.R.T., 2018. The Everglades National Park and Big Cypress National Preserve Vegetation Mapping Project: Interim report—Southwest Coastal Everglades (Region 3).
- Ruwaimana, M., Satyanarayana, B., Otero, V., M. Muslim, A., Syafiq A., M., Ibrahim, S., Raymaekers, D., Koedam, N., Dahdouh-Guebas, F., Joseph, S., 2018. The advantages of using drones over space-borne imagery in the mapping of mangrove forests. *PLoS ONE* 13 (7), e0200288. <https://doi.org/10.1371/journal.pone.0200288>.
- Schäfer, E., Heiskanen, J., Heikinheimo, V., Pellikka, P., 2016. Mapping tree species diversity of a tropical montane forest by unsupervised clustering of airborne imaging spectroscopy data. *Ecol. Ind.* 64, 49–58. <https://doi.org/10.1016/j.ecolind.2015.12.026>.
- Simard, M., Fatoyinbo, L., Smetanka, C., Rivera-Monroy, V.H., Castañeda-Moya, E., Thomas, N., Van der Stocken, T., 2019. Mangrove canopy height globally related to precipitation, temperature and cyclone frequency. *Nat. Geosci.* 12 (1), 40–45. <https://doi.org/10.1038/s41561-018-0279-1>.
- Singh, M., Evans, D., Tan, B.S., Nin, C.S., Loustau, D., 2015. Mapping and Characterizing Selected Canopy Tree Species at the Angkor World Heritage Site in Cambodia Using Aerial Data. *PLoS ONE* 10 (4), e0121558. <https://doi.org/10.1371/journal.pone.0121558>.
- Soares, M.L.G., Schaeffer-Novelli, Y., 2005. Above-ground biomass of mangrove species. I. Analysis of models. *Estuar. Coast. Shelf Sci.* 65 (1–2), 1–18. <https://doi.org/10.1016/j.ecss.2005.05.001>.
- Song, C., Dickinson, M.B., Su, L., Zhang, S.u., Yaussey, D., 2010. Estimating average tree crown size using spatial information from Ikonos and QuickBird images: Across-sensor and across-site comparisons. *Remote Sens. Environ.* 114 (5), 1099–1107. <https://doi.org/10.1016/j.rse.2009.12.022>.
- Spalding, M., Leal, M., 2021. The state of the worlds mangroves 2021, Global Mangrove Alliance.
- Story, M., Congalton, R.G., 1986. *Remote Sensing Brief Accuracy Assessment: A User's Perspective*. *Photogramm. Eng. Remote Sens.* 52, 397–399.
- Sun, W., Chen, B., Messinger, D.W., 2014. Nearest-neighbor diffusion-based pan-sharpening algorithm for spectral images. *Opt. Eng.* 53 (1), 013107. <https://doi.org/10.1117/1.OE.53.1.013107>.
- Taureau, F., Robin, M., Proisy, C., Fromard, F., Imbert, D., Debaine, F., 2019. Mapping the Mangrove Forest Canopy Using Spectral Unmixing of Very High Spatial Resolution Satellite Images. *Remote Sensing* 11, 367. <https://doi.org/10.3390/rs11030367>.
- Thomas, N., Lucas, R., Bunting, P., Hardy, A., Rosenqvist, A., Simard, M., Joseph, S., 2017. Distribution and drivers of global mangrove forest change, 1996–2010. *PLoS ONE* 12 (6), e0179302. <https://doi.org/10.1371/journal.pone.0179302>.
- Tochon, G., Férét, J.B., Valero, S., Martin, R.E., Knapp, D.E., Salembier, P., Chanasot, J., Asner, G.P., 2015. On the use of binary partition trees for the tree crown segmentation of tropical rainforest hyperspectral images. *Remote Sens. Environ.* 159, 318–331. <https://doi.org/10.1016/j.rse.2014.12.020>.
- Tong, F., Tong, H., Mishra, R., Zhang, Y., 2021. Delineation of Individual Tree Crowns Using High Spatial Resolution Multispectral WorldView-3 Satellite Imagery. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 14, 7751–7761. <https://doi.org/10.1109/JSTARS.2021.3100748>.
- Torresan, C., Berton, A., Carotenuto, F., Di Gennaro, S.F., Gioli, B., Matese, A., Miglietta, F., Vagnoli, C., Zaldei, A., Wallace, L., 2017. Forestry applications of UAVs in Europe: a review. *Int. J. Remote Sens.* 38 (8–10), 2427–2447. <https://doi.org/10.1080/01431161.2016.1252477>.
- Valderrama-Landeros, L., Flores-de-Santiago, F., Kovacs, J.M., Flores-Verdugo, F., 2018. An assessment of commonly employed satellite-based remote sensors for mapping mangrove species in Mexico using an NDVI-based classification scheme. *Environ. Monit. Assess.* 190 (1) <https://doi.org/10.1007/s10661-017-6399-z>.
- Vo, Q.T., Kuenzer, C., Vo, Q.M., Moder, F., Oppelt, N., 2012. Review of valuation methods for mangrove ecosystem services. *Ecol. Ind.* 23, 431–446. <https://doi.org/10.1016/j.ecolind.2012.04.022>.

- Vovides, A.G., Berger, U., Grueters, U., Guevara, R., Pommerening, A., Lara-Domínguez, A.L., López-Portillo, J., Larjavaara, M., 2018. Change in drivers of mangrove crown displacement along a salinity stress gradient. *Funct. Ecol.* 32 (12), 2753–2765. <https://doi.org/10.1111/1365-2435.13218>.
- Wagner, F.H., Ferreira, M.P., Sanchez, A., Hirye, M.C.M., Zortea, M., Gloor, E., Phillips, O.L., de Souza Filho, C.R., Shimabukuro, Y.E., Aragão, L.E.O.C., 2018. Individual tree crown delineation in a highly diverse tropical forest using very high resolution satellite images. *ISPRS J. Photogramm. Remote Sens.* 145, 362–377. <https://doi.org/10.1016/j.isprsjprs.2018.09.013>.
- Wagner, F.H., Sanchez, A., Tarabalka, Y., Lotte, R.G., Ferreira, M.P., Aïdar, M.P.M., Gloor, E., Phillips, O.L., Aragão, L.E.O.C., Pettorelli, N., Clerici, N., 2019. Using the U-net convolutional network to map forest types and disturbance in the Atlantic rainforest with very high resolution images. *Remote Sens. Ecol. Conserv.* 5 (4), 360–375. <https://doi.org/10.1002/rse2.111>.
- Wang, L., Gong, P., Biging, G.S., 2004. Individual Tree-Crown Delineation and Treetop Detection in High-Spatial-Resolution Aerial Imagery. *Photogramm. Eng. Remote Sens.* 70, 351–357. <https://doi.org/10.14358/PERS.70.3.351>.
- Wang, L.e., Jia, M., Yin, D., Tian, J., 2019. A review of remote sensing for mangrove forests: 1956–2018. *Remote Sens. Environ.* 231, 111223. <https://doi.org/10.1016/j.rse.2019.111223>.
- Wannasiri, W., Nagai, M., Honda, K., Santitamnont, P., Miphokasap, P., 2013. Extraction of Mangrove Biophysical Parameters Using Airborne LiDAR. *Remote Sensing* 5, 1787–1808. <https://doi.org/10.3390/rs5041787>.
- Ward, R.D., Friess, D.A., Day, R.H., Mackenzie, R.A., 2016. Impacts of climate change on mangrove ecosystems: a region by region overview. *Ecosyst. Health Sustainability* 2 (4), e01211. <https://doi.org/10.1002/ehs2.1211>.
- Xu, X., Zhou, Z., Tang, Y., Qu, Y., 2021. Individual tree crown detection from high spatial resolution imagery using a revised local maximum filtering. *Remote Sens. Environ.* 258, 112397. <https://doi.org/10.1016/j.rse.2021.112397>.
- Yin, D., Wang, L., 2019. Individual mangrove tree measurement using UAV-based LiDAR data: Possibilities and challenges. *Remote Sens. Environ.* 223, 34–49. <https://doi.org/10.1016/j.rse.2018.12.034>.
- Yin, D., Wang, L.e., 2016. How to assess the accuracy of the individual tree-based forest inventory derived from remotely sensed data: a review. *Int. J. Remote Sens.* 37 (19), 4521–4553. <https://doi.org/10.1080/01431161.2016.1214302>.
- Yu, K., Hao, Z., Post, C.J., Mikhailova, E.A., Lin, L., Zhao, G., Tian, S., Liu, J., 2022. Comparison of Classical Methods and Mask R-CNN for Automatic Tree Detection and Mapping Using UAV Imagery. *Remote Sensing* 14, 295. <https://doi.org/10.3390/rs14020295>.
- Yun, T., Jiang, K., Li, G., Eichhorn, M.P., Fan, J., Liu, F., Chen, B., An, F., Cao, L., 2021. Individual tree crown segmentation from airborne LiDAR data using a novel Gaussian filter and energy function minimization-based approach. *Remote Sens. Environ.* 256, 112307. <https://doi.org/10.1016/j.rse.2021.112307>.
- Zhang, K., 2008. Identification of gaps in mangrove forests with airborne LIDAR. *Remote Sens. Environ.* 112 (5), 2309–2325. <https://doi.org/10.1016/j.rse.2007.10.003>.